Geophysical Journal International

*Geophys. J. Int.* (2014) **197,** 1299–1320
Advance Access publication 2014 May 2

doi: 10.1093/gji/ggt487

# Secular variation of the middle and late Miocene geomagnetic field recorded by the Columbia River Basalt Group in Oregon, Idaho and Washington, USA

Ada R. Dominguez[1] and Rob Van der Voo[2]

[1]*Department of Geosciences, University of Arizona, Tucson, AZ* 85721*, USA. E-mail: adominguez@email.arizona.edu*
[2]*Department of Earth and Environmental Sciences, University of Michigan, Ann Arbor, MI* 48109-1005*, USA*

Accepted 2013 November 28. Received 2013 November 28; in original form 2013 June 4

## SUMMARY

This study of 118 discrete volcanic flows from the Columbia River Basalt Group is aimed to determine their distribution of geomagnetic field directions and virtual geomagnetic poles (VGPs) and to compare the inherent secular variation parameters with those from other studies. The magnetic signature of these rocks is uniformly carried by primary titanomagnetite, indicating that magnetic changes are due to variations in the magnetic field. Although most flows are flat lying, those that are tilted pass the Tauxe and Watson tilt test. Sequential flows with statistically similar site means were grouped, and directions that were considered outliers were evaluated and removed using the Vandamme cut-off method. Three normal-polarity (N-polarity) and three reversed-polarity (R-polarity) intervals are revealed by the stratigraphically ordered flows and have mean directions of N polarity (dec/inc = 6.6°/+61.2°, $k$ = 29.3, $\alpha_{95}$ = 4.2°), and R polarity (dec/inc = 178.2°/−59.2°, $k$ = 16, $\alpha_{95}$ = 5.5°). Regression analysis indicates that the secular variation analysis has not been affected by regional rotation, and that apparent polar wander is negligible. The VGP distribution is almost perfectly circular and supports the preference of VGP positions for the dispersion analysis. Dispersion parameters with corrections for within-site scatter ($S_b$) show a range of 14.3°–25.5°, including error limits, and were consistently higher for R-polarity results than for those of N polarity. Published dispersion parameters for extrusives <5 Ma show $S_b$ values slightly lower than ours, yielding values of 16°–19°, although the difference is not statistically significant. In contrast, published dispersion parameters from high quality data from the Cretaceous Normal Superchron are lower than those for the Neogene, which suggests that the noisiness of the magnetic field correlates with the frequency of reversals. Our new results allow us to extend the Plio-Pleistocene palaeosecular variation database to the bottom of the middle Miocene. Many Miocene formations on a variety of continents are suitable targets for future analysis. Furthermore, the significant difference between the reversed and N-polarity dispersion parameters is intriguing and needs substantiation.

**Key words:** Time series analysis; Palaeomagnetic secular variation; Reversals: process, time scale, magnetostratigraphy; North America.

## INTRODUCTION

The Earth's magnetic field is not constant, but undergoes rapid changes in orientation, as seen, for instance, by changing compass measurements over a decade. The adjustment to arrive at the direction to geographic north is done by accounting for the deviation in the horizontal plane and is called the declination correction. Inclination changes are equally present, even if less easily observed. Historical values of the inclinations at the Etna volcano show shifts of some 15° in the last 400 yr. The magnetic field strength also changes and variations of plus or minus 100 nT per year are known to occur, which is about 0.16 per cent of the total field. A corollary of the changing magnetic directions is that the north magnetic pole (where the field is vertically downward) changes its location, and is projected to have traversed some 9° in northern Canadian latitudes in the last 30 yr (i.e. since 1972; Olsen & Mandea 2007). The velocity of the moving north magnetic pole is thought to have increased in this interval from 20 to 55 km yr$^{-1}$. However, the south-dip pole has not shown similar acceleration (Olsen & Mandea 2007).

Collectively, these changes in intensity and orientation in the geological past are called palaeosecular variation (PSV) of the geomagnetic field, where the term secular implies that the changes

© The Authors 2014. Published by Oxford University Press on behalf of The Royal Astronomical Society.



GJI Geomagnetism, rock magnetism and palaeomagnetism

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

occur on a century-scale. As concisely stated by Linder & Gilder (2012), 'Secular variation . . . reflects the energy state of the geodynamo'. PSV can conveniently be expressed numerically as the time derivative of the so-called Gaussian coefficients, which describe the magnitudes of the infinite series of solutions to Laplace's equation in spherical harmonic analysis (SHA; Jacobs 1994). This mathematical treatment of the magnetic field imperfectly approximates the physics of the field, but it is clear that the energy state mentioned above does, in turn, reflect the convective vigor of the outer core (Linder & Gilder 2011). SHA is carried out in a reference frame with its coordinate origin at the Earth's centre and with alignment of the main dipole axis with the average position of the rotation axis. Palaeomagnetists have adopted this as a fundamental assumption in their geomagnetic models. The assumption is called the geocentric axial dipole hypothesis. For the present paper, the mathematical details of SHA are not important, because the Gaussian coefficients are only knowable for the magnetic field's configuration of the last several centuries. Studies of PSV, as the secular variation in geological time, are instead constrained to describe changes in the local directions of magnetization as a function of time. PSV includes the non-axial geocentric dipole as well as non-dipole contributions. Our investigation of PSV recorded by the Columbia River Basalts (CRB) aims to determine such changes in the local geomagnetic field in the mid to late Miocene.

Considering the rapidity of the changes, it is evident that these reflect the generating motions of the low-viscosity material in the liquid outer core, given that solid parts of the Earth's interior are highly unlikely to attain the speeds on the order of a millimetre per second that correspond to the previously mentioned rates of north magnetic pole migration. In contrast, the Earth's rotation axis, marked by its geographic north pole, is relatively stationary, ignoring wobbles with amplitudes in metres and very short periods on the order of up to 14 months, as well as long-period movements attributable to true polar wander. For the purposes of this paper, the rotation (or geographic) axis can be considered as very stable and stationary.

The geomagnetic field is generated by the complex interactions of (i) moving electrically conducting melts, (ii) magnetic fields with long- and short-term decay constants and (iii) electrical currents resulting from the interactions of the other two components. After time intervals of some 250 000 yr (on average, but highly variable), the field is known to have reversed itself. This statement typically excludes short excursions (Laj & Channell 2007), which are real but controversial in terms of whether they belong in the same category of geomagnetic phenomena as reversals. An actual reversal happens when a previous field, with a vertically down-dip pole in the Northern Hemisphere, decays and disappears in perhaps several thousand years. At the same time a new field builds up with a vertically up dip pole in the Northern Hemisphere, and with a staying power of some $10^5$–$10^8$ yr. The terminology of the polarity states is well known, but not necessarily logical, and includes 'normal' (N) polarity for that resembling today's and 'reversed' (R) polarity as it existed before 780 $\pm$ 10 ka at the end of the Matuyama Chron.

The question naturally arises as to how reversals are related to the energy state of the geodynamo. This question has motivated us to select the study area of the Miocene basalts in Oregon, Washington and Idaho, because it is an excellent target meeting many criteria for thorough PSV analysis (McElhinny & McFadden 1997).

Of particular interest to this study are the next three interrelated questions: (i) is there a correlation between a relatively low magnitude of secular variation and an enhanced palaeointensity of the field? or (ii) will the field be reversing more frequently, when it behaves 'noisily' (i.e. with larger-scale secular variation)? or (iii) will the field be more stable and show fewer and infrequent reversals when palaeointensity is high? Answers to these questions are anticipated to be affirmative (Biggin *et al.* 2008a, 2012), but it would be desirable to augment the evidence with substantive data. Question (ii) in particular will be addressed below.

Reversal rates are reasonably well known for the last 300 million years (Lowrie & Kent 2004) and more spottily so for the 540–300 Ma interval. A sustained lack of reversals is well known for the Permo-Carboniferous reversed superchron (PCRS) in Late Carboniferous to Early Permian times as well as for the Cretaceous Normal Superchron (CNS). Efforts to characterize secular variation in stratified igneous rocks formed during the CNS have been underway for a few years (see Biggin *et al.* 2008a). Results from sedimentary rocks of the PCRS (Choiniere & Swanson 1979; Haldan *et al.* 2009; Franco *et al.* 2012) have also been published, but the results remain scarce, or even ambiguous, given the possibility of inclination error (Kodama 2012). For the last 5 million years, a concerted effort (time-averaged field initiative or TAFI) has been underway involving many palaeomagnetists (see Johnson *et al.* 2008). Lastly, for the Precambrian, studies have been published by Biggin *et al.* (2008b), Tauxe & Kodama (2009) and Smirnov *et al.* (2011). It is evident that large gaps in the palaeomagnetic record exist, notably, dispersion studies are lacking for most of the Palaeozoic (Bazhenov *et al.* 2013).

Because the CRB group has been extensively studied for its palaeomagnetic directions, polarity records, age determinations and tectonic rotation patterns (Reidel 1978; Hooper & Conrey 1989; Wells *et al.* 1989; Reidel *et al.* 2003; Barry *et al.* 2010, and references therein), it presented itself as an obvious target for a PSV study, which led to the MSc thesis of the junior author (Dominguez 2012). Much of the tri-state area in Oregon, Washington and Idaho presents excellent exposures of mostly flat-lying basalts (Figs 1 and 2b), which were sampled in 2010. Our goal is, first of all, to determine whether the Miocene virtual geomagnetic poles (VGPs) are circularly distributed, as is predicted by most field models, and, in turn to determine whether the geomagnetic field directions recorded by the basalts have an elongated distribution. If the VGPs are circularly distributed, they are the favoured data to be analysed for PSV. This is not a foregone conclusion: recent studies have argued in favour of analysing directions instead of VGPs (Linder & Gilder 2011, 2012). Our longer term goals include studies in progress of Permian rocks from the PCRS (Menzo 2013; Bazhenov *et al.* 2014), for which no PSV records are yet available from igneous rocks.

The present study is for a time in the mid to late Miocene when reversals were about as frequent (5.2 per m.y.) as in the last 5 million year interval (4.2. per m.y.; Lowrie & Kent 1983), and this allows us to test whether reversal frequency relates to PSV magnitude in the same way in the Miocene as for the Plio-Pleistocene (Johnson *et al.* 2008). A different goal for the future, albeit at this time only barely realizable, is to explore whether PSV steadily changes in the very long term (see Smirnov *et al.* 2011; Biggin *et al.* 2012). A systematic trend may be interpreted as a consequence of the Earth's inner core growth, while the liquid outer core, responsible for the magnetic field, decreases in size or may reflect a changing heat flux through the core-mantle boundary.

A continued effort to obtain PSV data, not unlike those of TAFI, is possible in the future for Miocene time, and would allow expansion of the characterization to different latitudes. A quick survey of the database reveals many Miocene igneous rocks in a variety of continental units that have yielded promising palaeomagnetic results. The present study could then be seen as a kick-starter in this endeavour.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

**Table 1.** Palaeomagnetic site means, in stratigraphic order.

| Rank | Site | Member | | Dec | Inc | N1/N2 | k | $\alpha 95$ | GPS Lat. | GPS Long. | VGP Lat. | VGP Long. | N R | NRO | Combine? |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 15 | Saddle | Mntn. | 334.1 | 64.8 | 6/6 | 62.7 | 8.5 | 46.030 | −117.256 | 72.2 | 164.4 | | – | no |
| 2 | 118 | Saddle | Mntn. | 8.6 | 60.0 | 6/6 | 199.7 | 4.8 | 46.525 | −117.582 | 81.6 | 11.6 | | – | no |
| T** | *76* | *Saddle* | *Mntn.* | *336.0* | *−9.3* | *6/6* | *77.0* | *7.7* | *46.428* | *−117.169* | *−34.6* | *272.4* | | 0.939 | no |
| T** | *109* | *Saddle* | *Mntn.* | *325.8* | *−5.5* | *5/3* | *30.6* | *22.7* | *46.538* | *−117.560* | *−32.2* | *284.0* | | 0.939 | no |
| 3 | 12 | Saddle | Mntn. | 187.0 | −59.9 | 5/2 | *(81.8)* | *(28.0)* | 46.074 | −117.211 | −82.7 | 196.6 | | 0.939 | no |
| 4 | 4 | Saddle | Mntn. | 121.4 | −69.6 | 7/7 | 23.1 | 12.8 | 46.054 | −117.232 | −52.5 | 5.9 | | 0.939 | no |
| 5 | 46 | Saddle | Mntn. | 164.6 | −78.6 | 6/5 | 67.1 | 9.4 | 46.414 | −117.272 | −67.0 | 48.0 | | 0.939 | no |
| 6 | 42 | Saddle | Mntn. | 170.2 | −59.3 | 6/5 | 96.4 | 7.8 | 46.428 | −117.072 | −80.5 | 294.9 | | 0.939 | no |
| 7 | 37 | Saddle | Mntn. | 206.6 | −67.5 | 6/6 | 66.3 | 8.3 | 46.419 | −117.112 | −72.0 | 130.5 | | 0.939 | no |
| 8 | 41 | Saddle | Mntn. | 184.6 | −69.0 | 6/5 | 202.3 | 5.4 | 46.432 | −117.058 | −83.3 | 87.4 | | 0.939 | no |
| 9 | 74 | Saddle | Mntn. | 39.0 | 57.5 | 6/6 | 31.0 | 12.2 | 46.427 | −117.171 | 60.3 | 209.2 | | – | no |
| * | *75* | *Saddle* | *Mntn.* | *29.0* | *56.5* | *6/5* | *9.7* | *25.9* | *46.428* | *−117.170* | *66.6* | *220.3* | | – | no |
| 10 | 83 | Saddle | Mntn. | 29.7 | 46.5 | 7/7 | 41.2 | 9.5 | 46.518 | −117.794 | 60.1 | 236.2 | | – | no |
| 11 | 84 | Saddle | Mntn. | 21.3 | 48.8 | 6/6 | 36.7 | 11.2 | 46.516 | −117.790 | 66.4 | 245.7 | | – | no |
| ** | *68* | *Saddle* | *Mntn.* | *119.1* | *−53.0* | *5/3* | *66.3* | *15.3* | *46.469* | *−117.232* | *−42.8* | *339.6* | | 0.999 | yes |
| 12 | 62 | Saddle | Mntn. | 161.2 | −22.9 | 6/6 | 115.1 | 6.3 | 46.432 | −117.143 | −52.0 | 273.7 | | 0.999 | yes |
| ** | *78* | *Saddle* | *Mntn.* | *137.3* | *−22.8* | *5/4* | *64.1* | *11.6* | *46.429* | *−117.166* | *−40.2* | *303.1* | | 0.999 | yes |
| 13 | 32 | Saddle | Mntn. | 225.5 | −48.5 | 6/6 | 265.7 | 4.1 | 45.996 | −117.278 | −51.1 | 161.6 | | 0.999 | yes |
| ** | *115* | *Saddle* | *Mntn.* | *252.2* | *−51.4* | *6/4* | *1096.0* | *2.8* | *46.477* | *−117.768* | *−34.3* | *139.8* | | 0.999 | yes |
| 14 | Gp1 | Saddle | Mntn. | 205.9 | −69.0 | 12/10 | 44.2 | 7.3 | 46.078 | −117.200 | −72.9 | 121.2 | | 0.999 | yes |
| | 10 | Saddle | Mntn. | 204.7 | −71.8 | 6/5 | 26.0 | 15.3 | 46.079 | −117.196 | −71.5 | 109.0 | | 0.999 | yes |
| | 11 | Saddle | Mntn. | 202.9 | −66.4 | 6/5 | 105.3 | 7.5 | 46.078 | −117.203 | −74.3 | 134.3 | | 0.999 | yes |
| 15 | 25 | Saddle | Mntn. | 199.1 | −75.9 | 6/6 | 75.5 | 7.8 | 45.973 | −117.261 | −69.6 | 87.7 | | 0.999 | yes |
| 16 | 23 | Saddle | Mntn. | 217.1 | −70.1 | 6/6 | 295.9 | 3.9 | 45.973 | −117.263 | −65.1 | 119.9 | | 0.999 | yes |
| 17 | Gp2 | Saddle | Mntn. | 202.5 | −68.2 | 4/4 | 475.5 | 4.2 | 46.006 | −117.245 | −74.2 | 124.7 | | 0.999 | yes |
| | 30 | Saddle | Mntn. | 210.0 | −69.2 | 5/3 | 39.4 | 19.9 | 45.972 | −117.256 | −69.5 | 122.4 | | 0.999 | yes |
| | 22 | Saddle | Mntn. | 206.3 | −70.5 | 6/5 | 51.0 | 10.8 | 45.975 | −117.260 | −71.2 | 115.5 | | 0.999 | yes |
| | 13 | Saddle | Mntn. | 200.0 | −64.1 | 6/6 | 19.1 | 15.7 | 46.071 | −117.218 | −76.1 | 146.5 | | 0.999 | yes |
| | 5 | Saddle | Mntn. | 194.8 | −68.4 | 6/6 | 81.0 | 7.5 | 46.047 | −117.238 | −78.8 | 117.5 | | 0.999 | yes |
| 18 | 3 | Saddle | Mntn. | 216.9 | −68.8 | 7/5 | 43.5 | 11.7 | 46.324 | −117.110 | −65.5 | 125.4 | | 0.999 | yes |
| 19 | 2 | Saddle | Mntn. | 200.9 | −51.6 | 6/6 | 78.4 | 7.6 | 46.323 | −117.125 | −68.7 | 186.8 | | 0.999 | yes |
| 20 | 65 | Saddle | Mntn. | 170.7 | −32.8 | 6/6 | 1059.0 | 2.1 | 46.461 | −117.219 | −60.4 | 260.9 | | 0.999 | yes |
| 21 | 67 | Saddle | Mntn. | 172.1 | −41.8 | 6/6 | 52.4 | 9.3 | 46.467 | −117.230 | −66.7 | 261.3 | | 0.999 | yes |
| 22 | 38 | Saddle | Mntn. | 199.9 | −44.8 | 6/6 | 199.3 | 4.8 | 46.423 | −117.123 | −64.5 | 197.8 | | 0.999 | yes |
| 23 | 43 | Saddle | Mntn. | 189.5 | −47.1 | 6/4 | 78.8 | 10.4 | 46.423 | −117.126 | −70.4 | 217.3 | | 0.999 | yes |
| * | *96* | *Saddle* | *Mntn.* | *181.1* | *−81.4* | *4/4* | *7.7* | *35.5* | *46.599* | *−117.785* | *−63.4* | *62.9* | | 0.999 | yes |
| ** | *24* | *Saddle* | *Mntn.* | *202.2* | *82.2* | *5/5* | *62.5* | *9.8* | *45.973* | *−117.273* | *31.6* | *236.0* | | – | no |
| 24 | 80 | Wanapum | | 194.7 | −66.8 | 7/7 | 85.8 | 6.6 | 46.642 | −117.269 | −79.8 | 6.2 | | 0.4408 | no |
| 25 | 112 | Wanapum | | 197.1 | −8.9 | 5/5 | 19.7 | 17.7 | 46.480 | −117.576 | −45.5 | 217.7 | | 0.4408 | no |
| 26 | 73 | Wanapum | | 189.2 | −56.7 | 6/6 | 51.9 | 9.4 | 46.461 | −117.980 | −78.6 | 202.2 | | 0.4408 | no |
| 27 | 1 | Wanapum | | 187.6 | −64.0 | 7/7 | 67.4 | 7.4 | 46.314 | −117.117 | −84.7 | 30.9 | | 0.4408 | no |
| 28 | 77 | Wanapum | | 160.2 | −18.7 | 8/7 | 65.3 | 7.5 | 46.428 | −117.168 | −49.5 | 273.8 | | 0.4408 | no |
| 29 | 64 | Wanapum | | 173.4 | −34.1 | 6/6 | 24.9 | 13.7 | 46.453 | −117.208 | −61.7 | 130.5 | | 0.4408 | no |

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

**Table 1.** (*Continued.*)

| Rank | Site | Member | | Dec | Inc | N1/N2 | k | α95 | GPS Lat. | GPS Long. | VGP Lat. | VGP Long. | N | R | NRO | Combine? |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 30 | 63 | Wanapum | | 174.4 | −28.4 | 5/5 | 58.5 | 10.1 | 46.453 | −117.204 | −58.3 | 127.6 | | | 0.4408 | no |
| 31 | 89 | Wanapum | | 191.9 | −74.1 | 5/4 | 56.6 | 12.3 | 46.526 | −117.799 | −74.5 | 84.7 | | | 0.4408 | no |
| 32 | Gp3 | Wanapum | | 26.5 | 59.8 | 10/8 | 50.1 | 7.9 | 46.554 | −117.673 | 65.7 | 343.7 | | | | |
| | 108 | Wanapum | | 37.9 | 53.4 | 4/2 | *(116.6)* | *(23.3)* | 46.543 | −117.568 | 58.8 | 342.6 | | | 0.999 | yes |
| | 93 | Wanapum | | 22.5 | 59.5 | 6/6 | 43.6 | 10.3 | 46.565 | −117.778 | 72.6 | 345.1 | | | 0.999 | yes |
| ** | *52* | *Wanapum* | | *193.4* | *26.7* | *6/5* | *19.7* | *17.7* | *46.441* | *−117.438* | *28.3* | *47.8* | | | 0.999 | yes |
| ** | *33* | *Wanapum* | | *58.4* | *16.9* | *6/6* | *36.3* | *11.3* | *46.003* | *−117.278* | *27.9* | *350.4* | | | 0.999 | yes |
| 33 | 90 | Wanapum | | 357.1 | 51.3 | 8/8 | 36.9 | 9.2 | 46.544 | −117.775 | 75.3 | 71.9 | | | 0.999 | yes |
| 34 | 91 | Wanapum | | 20.0 | 57.9 | 7/7 | 65.7 | 7.5 | 46.559 | −117.772 | 73.3 | 353.7 | | | 0.999 | yes |
| 35 | 116 | Wanapum | | 22.9 | 67.8 | 6/6 | 25.9 | 13.4 | 46.572 | −117.738 | 74.4 | 308.3 | | | 0.999 | yes |
| 36 | 54 | Wanapum | | 359.4 | 41.9 | 6/6 | 163.8 | 5.2 | 46.463 | −117.461 | 67.7 | 64.0 | | | 0.999 | yes |
| 37 | 53 | Wanapum | | 289.2 | 80.8 | 6/6 | 37.8 | 11.0 | 46.445 | −117.442 | 49.4 | 216.0 | | | 0.999 | yes |
| 38 | 113 | Wanapum | | 342.0 | 56.1 | 6/6 | 58.7 | 8.8 | 46.647 | −117.723 | 73.3 | 122.0 | | | 0.999 | yes |
| 39 | Gp4 | Grande | Ronde | 327.3 | 71.6 | 10/7 | 24.2 | 12.5 | 46.004 | −117.263 | 76.4 | 213.9 | | | 0.999 | yes |
| | 26 | Grande | Ronde | 325.7 | 77.0 | 5/4 | 201.7 | 6.5 | 45.974 | −117.271 | 63.3 | 211.0 | | | 0.999 | yes |
| | 14 | Grande | Ronde | 342.9 | 75.0 | 5/3 | 81.5 | 13.7 | 46.034 | −117.255 | 71.4 | 216.9 | | | 0.999 | yes |
| 40 | 70 | Grande | Ronde | 0.4 | 74.6 | 6/6 | 64.1 | 8.4 | 46.626 | −117.339 | 75.5 | 243.4 | | | 0.999 | yes |
| 41 | 117 | Grande | Ronde | 337.2 | 70.9 | 6/6 | 70.1 | 8.1 | 46.551 | −117.677 | 73.3 | 192.2 | | | 0.999 | yes |
| 42 | 17 | Grande | Ronde | 305.4 | 72.0 | 6/6 | 61.2 | 8.6 | 46.003 | −117.280 | 55.3 | 191.4 | | | 0.999 | yes |
| 43 | 8 | Grande | Ronde | 332.7 | 74.7 | 6/6 | 89.2 | 7.1 | 46.043 | −117.253 | 68.1 | 206.6 | | | 0.999 | yes |
| 44 | 7 | Grande | Ronde | 359.3 | 59.8 | 6/6 | 79.9 | 7.5 | 46.056 | −117.239 | 84.6 | 68.5 | | | 0.999 | yes |
| 45 | 92 | Grande | Ronde | 27.2 | 56.9 | 4/4 | 69.6 | 11.1 | 46.563 | −117.775 | 68.0 | 346.8 | | | 0.999 | yes |
| 46 | Gp5 | Grande | ronde | 19.7 | 76.5 | 3/3 | 156.0 | 9.9 | 46.179 | −117.457 | 68.5 | 265.9 | | | 0.999 | yes |
| | 18 | Grande | Ronde | 27.1 | 74.3 | 6/6 | 28.8 | 12.7 | 46.005 | −117.277 | 68.5 | 280.1 | | | 0.999 | yes |
| | 16 | Grande | Ronde | 20.7 | 83.4 | 6/6 | 17.4 | 16.5 | 46.010 | −117.275 | 57.9 | 251.4 | | | 0.999 | yes |
| | 82 | Grande | Ronde | 13.0 | 71.5 | 3/3 | 44.0 | 18.8 | 46.523 | −117.818 | 77.4 | 277.2 | | | 0.999 | yes |
| 47 | 88 | Grande | Ronde | 4.1 | 56.5 | 5/5 | 105.3 | 7.5 | 46.526 | −117.804 | 80.1 | 43.0 | | | 0.999 | yes |
| ** | *34* | *Grande* | *Ronde* | *263.7* | *73.4* | *6/5* | *61.8* | *9.8* | *46.002* | *−117.279* | *35.4* | *204.1* | | | 0.999 | yes |
| 48 | Gp6 | Grande | Ronde | 4.0 | 54.3 | 12/12 | 60.5 | 5.6 | 46.005 | −117.519 | 78.9 | 40.1 | | | 0.999 | yes |
| | 35 | Grande | Ronde | 8.2 | 54.2 | 6/6 | 30.1 | 12.4 | 46.009 | −117.277 | 77.1 | 31.0 | | | 0.999 | yes |
| | 114 | Grande | Ronde | 2.7 | 56.8 | 6/6 | 497.9 | 3.0 | 46.640 | −117.761 | 80.5 | 49.2 | | | 0.999 | yes |
| 49 | 55 | Grande | Ronde | 351.6 | 49.9 | 5/4 | 517.4 | 4.0 | 46.447 | −117.464 | 73.0 | 88.0 | | | 0.999 | yes |
| 50 | 60 | Grande | Ronde | 3.2 | 60.2 | 6/6 | 120.6 | 5.5 | 46.468 | −117.506 | 82.6 | 43.1 | | | 0.999 | yes |
| 51 | Gp7 | Grande | Ronde | 353.3 | 54.2 | 13/12 | 389.5 | 6.3 | 46.537 | −117.688 | 77.2 | 88.3 | | | 0.999 | yes |
| | 61 | Grande | Ronde | 349.9 | 53.5 | 6/5 | 72.5 | 9.0 | 46.478 | −117.646 | 75.4 | 97.6 | | | 0.999 | yes |
| | 81 | Grande | Ronde | 357.9 | 51.2 | 7/7 | 17.2 | 15.0 | 46.522 | −117.814 | 75.3 | 69.3 | | | 0.999 | yes |
| | 104 | Grande | Ronde | 351.6 | 57.6 | 6/5 | 64.1 | 9.6 | 46.598 | −117.603 | 79.6 | 101.9 | | | 0.999 | yes |
| 52 | Gp8 | Grande | Ronde | 355.8 | 68.0 | 10/6 | 17.7 | 16.4 | 46.503 | −117.474 | 80.4 | 149.3 | | | 0.999 | yes |
| | 49 | Grande | Ronde | 340.2 | 64.2 | 5/4 | 45.5 | 13.8 | 46.442 | −117.385 | 76.3 | 157.8 | | | 0.999 | yes |
| | 107 | Grande | Ronde | 352.4 | 63.6 | 5/2 | *(128.5)* | *(22.2)* | 46.563 | −117.563 | 84.5 | 140.8 | | | 0.999 | yes |
| 53 | 99 | Grande | Ronde | 339.4 | 69.7 | 8/5 | 76.2 | 8.8 | 46.656 | −117.806 | 75.2 | 187.3 | | | 0.999 | yes |
| 54 | 59 | Grande | Ronde | 12.0 | 26.0 | 6/6 | 254.8 | 4.2 | 46.468 | −117.493 | 55.7 | 41.5 | | | 0.999 | yes |
| 55 | 50 | Grande | Ronde | 358.9 | 48.8 | 6/5 | 460.2 | 3.6 | 46.442 | −117.392 | 73.3 | 66.0 | | | 0.999 | yes |
| 56 | Gp9 | Grande | Ronde | 359.8 | 43.9 | 5/5 | 844.5 | 2.6 | 46.493 | −117.467 | 69.2 | 62.9 | | | 0.999 | yes |

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

**Table 1.** (*Continued.*)

| Rank | Site | Member | | Dec | Inc | N1/N2 | k | α95 | GPS Lat. | GPS Long. | VGP Lat. | VGP Long. | N | R | NRO | Combine? |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 56 | Grande | Ronde | 358.9 | 42.5 | 6/5 | 69.3 | 9.3 | 46.447 | −117.467 | 68.2 | 65.3 | | | 0.999 | yes |
| | 45 | Grande | Ronde | 3.7 | 45.5 | 6/3 | 5312.0 | 3.4 | 46.408 | −117.264 | 70.3 | 52.9 | | | 0.999 | yes |
| | 44 | Grande | Ronde | 358.9 | 43.7 | 6/6 | 912.4 | 2.2 | 46.406 | −117.256 | 69.1 | 65.5 | | | 0.999 | yes |
| | 105 | Grande | Ronde | 2.7 | 44.8 | 5/5 | 37.8 | 12.6 | 46.589 | −117.579 | 69.7 | 55.5 | | | 0.999 | yes |
| | 101 | Grande | Ronde | 355.2 | 42.6 | 7/5 | 26.3 | 15.2 | 46.616 | −117.767 | 67.7 | 73.9 | | | 0.999 | yes |
| 57 | Gp10 | Grande | Ronde | 348.0 | 56.2 | 13/7 | 44.4 | 9.2 | 46.622 | −117.795 | 75.8 | 106.0 | | | 0.999 | yes |
| | 95 | Grande | Ronde | 341.8 | 56.2 | 8/3 | 29.6 | 23.1 | 46.594 | −117.782 | 73.3 | 122.7 | | | 0.999 | yes |
| | 98 | Grande | Ronde | 353.5 | 55.5 | 5/4 | 48.2 | 13.4 | 46.649 | −117.808 | 78.3 | 89.2 | | | 0.999 | yes |
| 58 | 58 | Grande | Ronde | 10.5 | 56.2 | 6/6 | 82.5 | 7.4 | 46.468 | −117.494 | 77.5 | 20.0 | | | 0.999 | yes |
| 59 | Gp11 | Grande | Ronde | 23.0 | 53.9 | 11/9 | 25.8 | 10.3 | 46.592 | −117.810 | 66.3 | 353.6 | | | 0.999 | yes |
| | 86 | Grande | Ronde | 25.2 | 53.2 | 7/7 | 19.8 | 13.9 | 46.524 | −117.819 | 67.1 | 357.0 | | | 0.999 | yes |
| | 100 | Grande | Ronde | 28.7 | 54.6 | 4/2 | *(52.3)* | *(35.2)* | 46.660 | −117.802 | 65.6 | 350.1 | | | 0.999 | yes |
| 60 | 103 | Grande | Ronde | 18.0 | 59.4 | 6/6 | 91.8 | 7.0 | 46.622 | −117.632 | 75.5 | 352.1 | | | 0.999 | yes |
| 61 | Gp12 | Grande | Ronde | 10.5 | 64.8 | 13/13 | 38.2 | 6.8 | 46.273 | −117.423 | 82.6 | 316.3 | | | 0.999 | yes |
| | 106 | Grande | Ronde | 6.7 | 67.0 | 8/8 | 31.4 | 10.0 | 46.577 | −117.571 | 84.6 | 295.2 | | | 0.999 | yes |
| | 28 | Grande | Ronde | 13.0 | 63.0 | 5/5 | 84.3 | 8.4 | 45.968 | −117.275 | 80.7 | 337.4 | | | 0.999 | yes |
| 62 | Gp13 | Grande | Ronde | 18.8 | 65.6 | 14/14 | 25.8 | 8.0 | 46.490 | −117.645 | 74.3 | 320.4 | | | 0.999 | yes |
| | 57 | Grande | Ronde | 24.4 | 61.0 | 6/6 | 51.9 | 9.4 | 46.456 | −117.472 | 72.1 | 337.9 | | | 0.999 | yes |
| | 87 | Grande | Ronde | 19.2 | 68.4 | 8/8 | 22.0 | 12.1 | 46.523 | −117.817 | 76.5 | 303.0 | | | 0.999 | yes |
| 63 | 94 | Grande | Ronde | 11.8 | 68.0 | 8/5 | 45.3 | 11.5 | 46.590 | −117.784 | 81.1 | 297.9 | | | 0.999 | yes |
| 64 | 51 | Grande | Ronde | 24.1 | 57.9 | 6/6 | 28.3 | 12.8 | 46.442 | −117.394 | 70.7 | 347.9 | | | 0.999 | yes |
| 65 | 110 | Grande | Ronde | 55.9 | 60.5 | 4/2 | *(89.0)* | *(26.8)* | 46.532 | −117.559 | 50.3 | 318.8 | | | 0.999 | yes |
| 66 | 102 | Grande | Ronde | 36.5 | 67.7 | 6/6 | 32.7 | 11.9 | 46.628 | −117.721 | 65.8 | 309.2 | | | 0.999 | yes |
| 67 | 9 | Grande | Ronde | 215.5 | −65.9 | 6/6 | 42.4 | 10.4 | 46.082 | −117.184 | −66.0 | 134.8 | | | 0.8726 | no |
| 68 | 29 | Grande | Ronde | 201.4 | −82.0 | 6/6 | 70.1 | 8.1 | 45.973 | −117.264 | −60.1 | 74.2 | | | 0.8726 | no |
| 69 | 27 | Grande | Ronde | 154.0 | −43.5 | 5/5 | 24.4 | 15.8 | 45.636 | −117.269 | −60.9 | 297.4 | | | 0.8726 | no |
| 70 | 21 | Grande | Ronde | 180.6 | −73.6 | 6/6 | 98.2 | 6.8 | 45.979 | −117.267 | −76.5 | 298.5 | | | 0.8726 | no |
| 71 | 31 | Grande | Ronde | 168.8 | −75.9 | 6/5 | 282.0 | 4.6 | 45.980 | −117.272 | −71.5 | 281.3 | | | 0.8726 | no |
| 72 | 20 | Grande | Ronde | 173.1 | −70.0 | 4/4 | 170.3 | 7.1 | 45.982 | −117.273 | −80.9 | 270.7 | | | 0.8726 | no |
| 73 | 19 | Grande | Ronde | 154.9 | −73.8 | 6/6 | 143.5 | 5.6 | 45.985 | −117.275 | −69.7 | 24.8 | | | 0.8726 | no |
| 74 | 48 | Grande | Ronde | 158.5 | −54.9 | 6/6 | 475.4 | 3.1 | 46.442 | −117.381 | −70.5 | 306.0 | | | 0.8726 | no |
| 75 | 72 | Grande | Ronde | 139.3 | −63.6 | 6/2 | *(164.1)* | *(19.6)* | 46.610 | −117.303 | −62.0 | 344.9 | | | 0.8726 | no |
| 76 | 47 | Grande | Ronde | 162.0 | −62.5 | 6/5 | 177.3 | 5.8 | 46.440 | −117.353 | −77.1 | 327.6 | | | 0.8726 | no |
| 77 | 6 | Grande | Ronde | 175.5 | −55.2 | 6/6 | 55.9 | 9.0 | 46.050 | −117.239 | −79.1 | 262.6 | | | 0.8726 | no |
| ** | *71* | *Grande* | *Ronde* | *4.0* | *−74.4* | *4/4* | *236.4* | *6.0* | *46.616* | *−117.317* | *−17.5* | *60.6* | | | 0.8726 | no |
| 78 | 69 | Grande | Ronde | 140.1 | −57.7 | 6/5 | 90.5 | 8.1 | 46.510 | −117.233 | −59.8 | 331.7 | | | 0.8726 | no |
| 79 | 79 | Grande | Ronde | 166.6 | −58.2 | 6/5 | 70.0 | 9.2 | 46.513 | −117.235 | −77.6 | 174.2 | | | 0.8726 | no |
| 80 | 66 | Grande | Ronde | 158.7 | −57.1 | 6/6 | 89.9 | 7.1 | 46.467 | −117.230 | −72.0 | 185.8 | | | 0.8726 | no |
| 81 | 36 | Grande | Ronde | 166.9 | −48.0 | 6/6 | 45.5 | 10.0 | 46.417 | −117.103 | −69.8 | 278.0 | | | 0.8726 | no |
| 82 | 39 | Grande | Ronde | 182.4 | −72.2 | 6/5 | 153.6 | 6.2 | 46.427 | −117.135 | −79.0 | 303.9 | | | 0.8726 | no |
| 83 | 40 | Grande | Ronde | 175.8 | −71.4 | 6/6 | 87.4 | 7.2 | 46.428 | −117.142 | −80.0 | 283.4 | | | 0.8726 | no |
| 84 | 111 | Grande | Ronde | 168.6 | −36.0 | 8/3 | 80.2 | 13.9 | 46.508 | −117.566 | −61.9 | 265.7 | | | 0.8726 | no |

Sites not used are in italics and include 10 beyond the Vandamme cut-off and two site means (75, 96) where $k < 11$.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

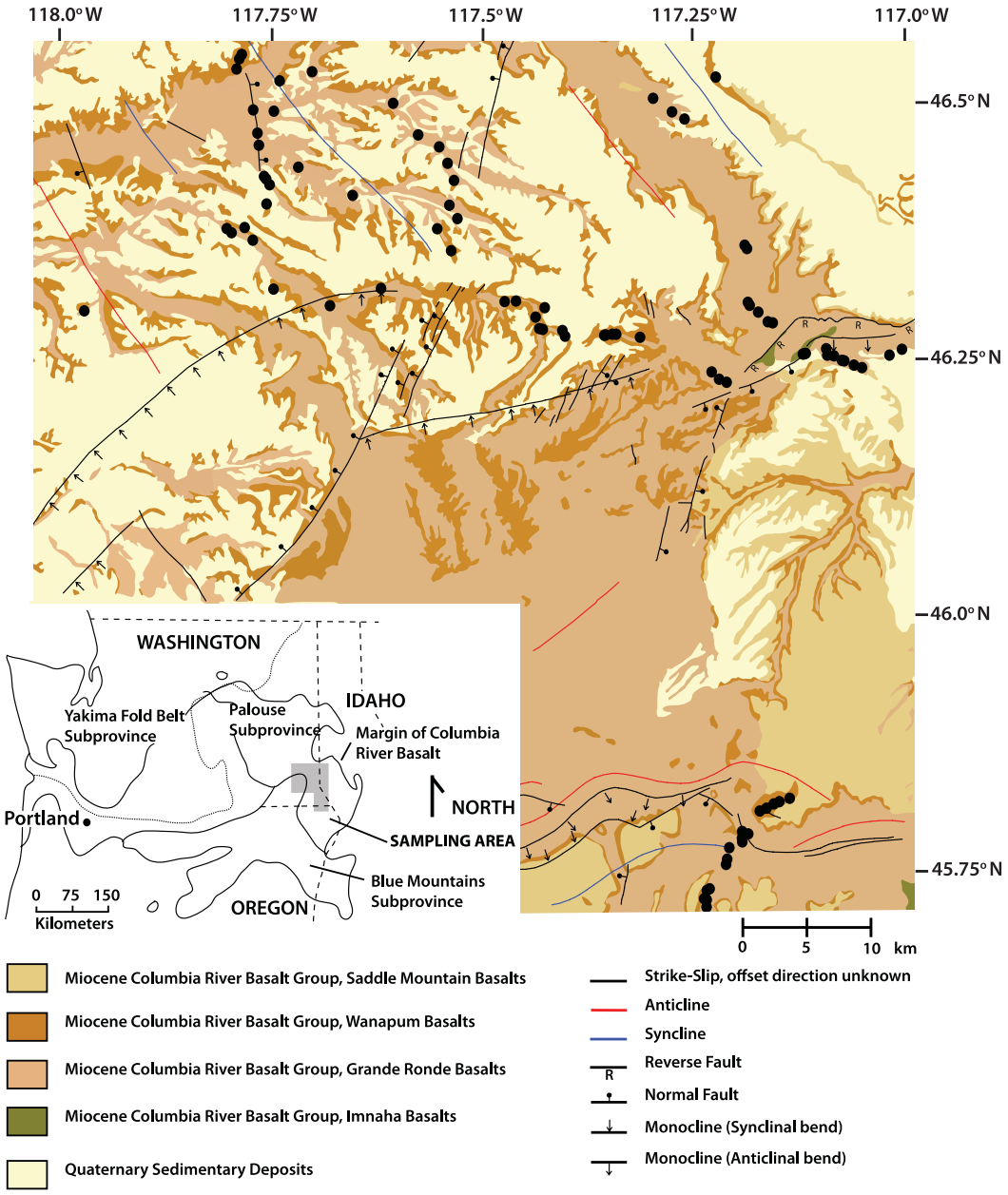



**Figure 1.** Map of sampled region, differentiated by basalt stratigraphy. The black circles indicate the sampled sites. The inset on the lower left shows the location of the sampled area [adapted from Hooper (2000), and the Washington Interactive Geologic Map (https://fortress.wa.gov/dnr/geology/)].

## GEOLOGICAL SETTING AND SAMPLING

The CRB, overlying the Columbia Plateau, are located between the volcanic subduction-related arc of the Cascade Range and the Rocky Mountains. They were erupted from linear fissures 17–6 Ma, as determined by K–Ar and $^{40}$Ar–$^{39}$Ar dating methods (Waters 1961; Watkins & Baksi 1974; McKee *et al.* 1977; Choiniere & Swanson 1979; Swanson *et al.* 1979a; 1981; Long & Duncan 1983; Tolan *et al.* 2009; Barry *et al.* 2010) on flows and dikes in parts of Washington, Oregon and Idaho. Most of the group was erupted between 17 and 15.5 Ma (Mangan *et al.* 1986). The flows form a thick basaltic sequence, consisting of at least 300 individual tholeiitic cooling units (Swanson *et al.* 1979b; Tolan *et al.* 1989). These flood basalts cover the area behind the Cascades Arc and expanded eastward, covering the Columbia Plateau, where some sections have basalt layers greater than 3 km thick (e.g. Reidel *et al.* 1989).

The origin of the CRB volcanism has been debated for decades. Given its nature as a large igneous province, a mantle plume origin has logically been considered (see Long *et al.* 2012, and references therein). However, the location behind the Cascade Volcanic Arc has led others (e.g. Atwater & Stock 1998; Druken *et al.* 2011) to adopt a backarc/rollback model for the volcanism. And as an illustration of the province's ambiguity, Chamberlain & Lambert (1994) called it 'a plume reinforced backarc magmatism' on the basis of lead isotope signatures. The Snake River Plain and Yellowstone volcanism (Fouch 2012) shows a spatial pattern of NE-ward diminishing ages, which resembles that of plume tracks, but

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

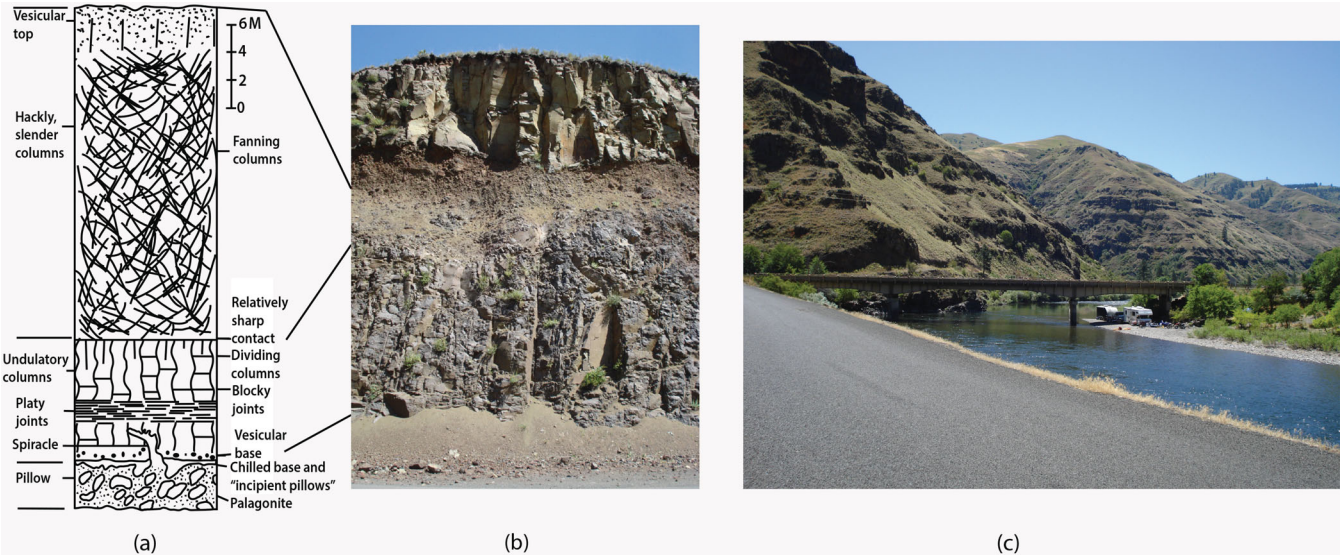



**Figure 2.** Visual representation of the Columbia River Basalt flows, showing (a) a schematic cross-section of an individual flow, (b) a field photograph showing the characteristics expected for the flows in this region and (c) horizontal flows near bridge over the Grande Ronde River.

even if true, West- and NW-ward continuation of the age patterns is difficult to discern, rendering the plume model unlikely for the CRB.

Palaeomagnetic studies of the CRB have shown that there has been a 16° regional clockwise rotation in western Washington since 12 Ma (Simpson *et al.* 1980; Magill *et al.* 1982) that is inferred to be due to tectonic dextral shearing between the westernmost section of the North American Plate and the northward motion of the Pacific Plate (Beck 1976; Beck 1980; Sheriff & Bentley 1980). However, the boundary between the stable and rotated sections occurs well to the east of the Cascades (Simpson *et al.* 1980). The Steens Mountain area and surroundings in central Oregon suffered a $14.5^{o} \pm 5.4^{o}$ clockwise rotation with respect to a global reference palaeopole (Jarboe *et al.* 2008). Eastern Washington and northeastern Oregon, however, are more stable, having undergone less than a 3° clockwise rotation (Magill *et al.* 1982).

Thus, to avoid rotation bias among our sites, only the eastern, stable part of the plateau was sampled in this study (Table 1; Fig. 1, east of 118°W). Following Swanson *et al.* (1979a,b), the CRB are divided into five formations (the Imnaha, Grande Ronde, Wanapum, Saddle Mountain and Picture Gorge Basalts), with the Grande Ronde Basalt group containing 85 per cent of the total volume (Reidel *et al.* 1981), and the Saddle Mountain Basalt composing less than 1 per cent of the total volume.

The area sampled in our study had very distinct exposures of individual flows. Each discrete flow had a characteristic arrangement (Fig. 2) of a vesicular or scoriaceous top, underlain by slender columns, undulatory columns, platy joints, spiracles and pillow basalts (Reidel 1983; Mangan *et al.* 1986). A saprolite layer of approximately less than 100 000 yr in duration is located between the Grande Ronde and Wanapum section in most places.

Each flow is a discrete site. The Grande Ronde (64 flows), Wanapum (18 flows) and Saddle Mountain (36 flows) formations could be sampled in stratigraphical order, and yielded 636 individually oriented core samples. The Imnaha and Picture Gorge formations were not sampled.

A compass and inclinometer were used to determine bedding attitudes, as well as azimuth and plunge of the core-samples. Solar compass readings were taken in (generally favourable) weather conditions. It turns out that magnetic field strengths at the outcrops were not high enough to deflect a magnetic compass needle significantly. The tops of flow ridges were not sampled so as to avoid lightning-induced remagnetization. When collecting the samples in the field, precautions were taken to avoid sampling the same flow unit twice. Most of the samples were collected in road-cuts due to difficulty in obtaining permission to sample on private property, which makes up the majority of the region. However, roads cut both horizontally and vertically through the sequences, so the individual units could be easily sampled and tracked.

The thickness of each flow was considerable (∼30 m thick) with sampling concentrated in the lower undulatory columns, overlain by the buffering upper vesicular section. This reduced the chances of sampling from an undesirable baked contact owing to an overlying flow. Thus, we can be confident that the magnetic measurements from an individual flow belong to that flow and not the one overlying it. Furthermore, the cooling time of the flows was calculated as about 7 yr according to the heat conduction equation, which is considered negligible. Granted that the magnetic mineralogy and its magnetization is primary, this warrants that the magnetic values for each flow were recorded soon after emplacement as spot readings of the geomagnetic field.

The magnetic declination correction for the region was made using the value (15°11′E) provided by the National Geophysical Data Center, for latitude 46°N and longitude 117°W during the 2010 July sampling season. Declinations, azimuths and strikes have all been corrected for this declination deviation.

## LABORATORY METHODS

In the laboratory, cores were cut into 2.1-cm long specimens and any visible rust stains from drill or saw were removed using coarse sandpaper. Samples were labelled with non-magnetic, temperature-resistant paint, and broken specimens were cemented with non-magnetic alumina cement.

All samples were stored and measured in a magnetically shielded room in the Paleomagnetism, Structure and Tectonics Laboratory at the University of Michigan to prevent viscous magnetizations

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

from accumulating in the samples. The natural remanent magnetization (NRM) of the rocks was measured using a three-axis 2G superconducting magnetometer. Most specimens were thermally demagnetized using an ASC TD-48 demagnetizer located in the shielded room where the residual field is generally less than 200 nT. Alternating field demagnetization was carried out on pilot samples (e.g. Figs 3a and b) and was attempted for some samples whose components were not successfully isolated with thermal demagnetization. Results of the demagnetization treatments have been graphed in orthogonal vector endpoint diagrams (Zijderveld 1967) and in stereographic projections. For calculation of the magnetization directions, the principal component analysis (Kirschvink 1980) was used on linear segments of the Zijderveld diagrams; in cases where stable endpoints were not obtained, as

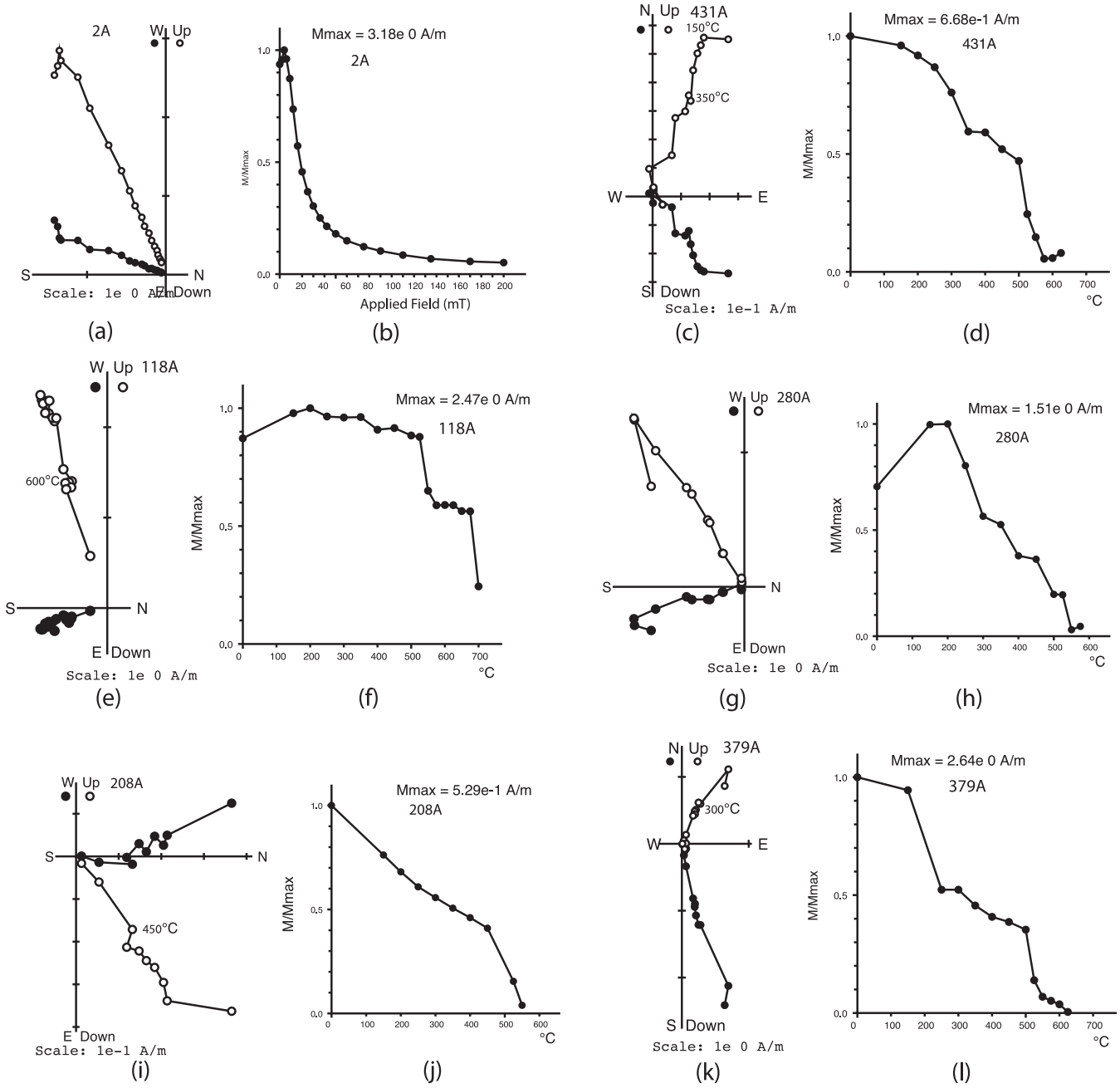



**Figure 3.** Orthogonal demagnetization diagrams (Zijderveld 1967) and intensity diagrams. (a) Sample CR2A (site 1), with clear univectorial decay, and (b) its smooth demagnetization pattern in alternating fields. (c) Sample CR431A (site 72) revealing a difference in direction between lower and higher temperature components and (d) its corresponding intensity diagram, which shows a corresponding decay between 100 and 350 °C and between 500 and 580 °C. (e) Sample CR118A (site 20) revealing a nearly univectorial SSE/up magnetization, and (f) its intensity diagram that depicts the presence of magnetite as well as uncommon hematite. (g) Sample CR280A (site 47) shows an overall southerly and up direction, while the intensity diagram (h) reveals distributed unblocking up to 580 °C. Sample CR208A (site 35) shows (i) a northerly and down direction above 450 °C, while (j) the intensity diagram indicates the presence of magnetite. (k) Sample CR379A (site 64) shows a curved trajectory in the Zijderveld diagram up to ∼300 °C, while its intensity diagram (l) indicates two unblocking ranges; the higher one (500–550 °C) indicating low-Ti magnetite, and a tail end above 600 °C, indicating a minor presence of hematite.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

identified by trajectories along great circle paths, a combined analysis of remagnetization circles and stable-endpoint observations (McFadden & McElhinny 1988) was used. The palaeomagnetic measurements were analysed with the use of the Paleomac software (Cogné 2003).

In order to image the most likely carriers of the magnetization of the samples, a scanning electron microscope (SEM; Hitachi S-570, Japan) fitted with an EDS analysis system was used to qualitatively assess their compositions and determine whether the magnetization is likely to be primary and carried by stable grains. The magnetic properties of the samples were measured at the Institute for Rock Magnetism in the University of Minnesota. Remanence curves, hysteresis loops and first-order reversal curves (FORC) were measured in high-, low- and room-temperature with applied fields using the Princeton Measurements micro-Vibrating Sample Magnetometers. For this instrument, hot (∼300–1025 K; 0–1.7 T), cold (10–473 K; 0–1.7 T) and room-temperature (∼300 K; 0–1.4 T) ranges were employed.

## RESULTS

### Unblocking characteristics

Alternating field and thermally demagnetized samples primarily show univectorial decay to the origin (Fig. 3). The moment ($M/M_{max}$) diagrams reveal a Curie point of 500–580 °C (Figs 3d, h and j), indicating that low-Ti magnetite is likely to be the main carrier of the NRM. Some samples showed a prominent decrease in magnetic intensity between 100 and 300 °C (Figs 3d, h and l); given that the corresponding vector (Figs 3c, g and k) in the Zijderveld diagrams resembles the characteristic remanent direction, it is unlikely that this lower temperature component represents anything but an ancient characteristic remanence, carried by a titanomagnetite (TM) or perhaps an iron-sulfide, although there are no other magnetic indications for greigite or pyrrhotite. This lower temperature direction is ignored since it is probably not as well isolated as the higher temperature component. Some samples have unblocking temperatures up to about 670 °C (Fig. 3f), indicative of an occasional coexistence of magnetite and hematite. This is perhaps the result of syn-depositional oxidation, because the hematite has the same direction as the magnetite components (Fig. 3e), so the hematite component is most likely acquired early.

Only two sites (85 and 97) did not yield any coherent or interpretable patterns in the Zijderveld plots and are not included in the tables or figures of this study. The other 116 sites are included in Table 1 from which they may have been selected for further analysis, as noted.

### Scanning electron microscopic analysis

In order to determine whether the magnetic directions could have been affected by alteration of the magnetic mineralogy, samples were examined by SEM and energy dispersive spectroscopy (EDS). Three representative samples were selected depending on the behaviour of their directions and magnetic moment (CR280B, CR208B and CR379C). Their characteristics are as follows:

*Sample CR280* appears to contain a low-titanium magnetite, with an unblocking temperature of ∼550 °C, accompanied by a low temperature component (unblocking between 200 and 300 °C) perhaps suggesting the presence of a titanomagnetite component (Figs 3g and h). The titanomagnetite and low-titanium magnetite components have nearly the same directions, which might indicate that their magnetizations are of nearly the same age. It is also possible that a present day field direction is hidden in a composite remanence removed between 200 and 300 °C.

*Sample CR208* yielded Zijderveld and intensity diagrams for this sample that show different behaviours depending on the demagnetization step (Figs 3i and j). From 150 to 450 °C, there is an unblocking behaviour that could be indicative of titanomagnetite, while another drop in intensity occurs between 450 and 550 °C, possibly due to the presence of low-titanium magnetite. The titanomagnetite and low-titanium magnetite components have different declinations, which might indicate that they are of different ages in this sample.

*Sample CR379* revealed three magnetic components: a low temperature component (unblocking temperature of 250 °C), a magnetite component (550 °C) and a hematite component (625 °C; Figs 3k and l).

The most significant SEM observation is that TM grains of 50–100 μm in size are seen to be exsolved (Fig. 4). As this exsolution is generally attributed to a high-temperature process, its presence supports the interpretation that the magnetic moments of the low-Ti magnetites date back to the cooling of the flow, and that the much reduced dimensions of the elongated lamellae make it possible for the magnetization to be more stable as pseudo-single-domain (PSD) instead of multi-domain (MD). The SEM images of exsolved low-Ti and high-Ti magnetite are in agreement with the rock magnetic unblocking patterns. Further observations about the SEM analyses can be found in Dominguez (2012).

### Rock magnetic measurements

*Thermomagnetic runs*. Magnetic susceptibility as a function of temperature is shown for two typical samples in Fig. 5. One sample shows a high Curie temperature, about 540 °C, indicative of low-Ti magnetite, with a reversible curve upon cooling. The other type has a Curie temperature of about 200 °C, indicative of TM (∼60), and is metastable, so that the cooling curve is much different from the heating curve.

*Hysteresis curves and Day diagram.* Representative hysteresis curves are shown in Fig. 6, and their parameters as moment versus coercivity ratios are plotted in Fig. 7, in a Day diagram (Day *et al.* 1977). A large majority of samples shows parameters in the PSD field, with a tendency to streak into the MD field (Fig. 7). PSD characteristics are also apparent in the FORC diagrams (Fig. 8a), with an occasional portrayal of a more single-domain like field (Fig. 8b).

*Low-temperature behaviour*. Cooling of a room-temperature saturation magnetization (Fig. 9) to less than 50 K shows a loss of about 15 per cent, a significant portion of which (4 per cent) is indicative of the Verwey transition at ∼125 K, when magnetite undergoes a transition from cubic to monoclinic crystal structure. The Verwey transition is a unique characteristic of PSD and MD magnetite, and is therefore in support of our overall interpretation of the rock magnetic data, namely that the carrier of the remanence is typically low-Ti magnetite, seen in several samples to be one end-member of a high-temperature exsolution process, and of PSD nature.

### Field tests and other reliability criteria

*Tilt test.* More than 85 per cent of the sampled sites have bedding attitudes that are essentially flat lying. Fifteen sites have non-zero dips, most likely due to draping over subsurface faults, of 10° or

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

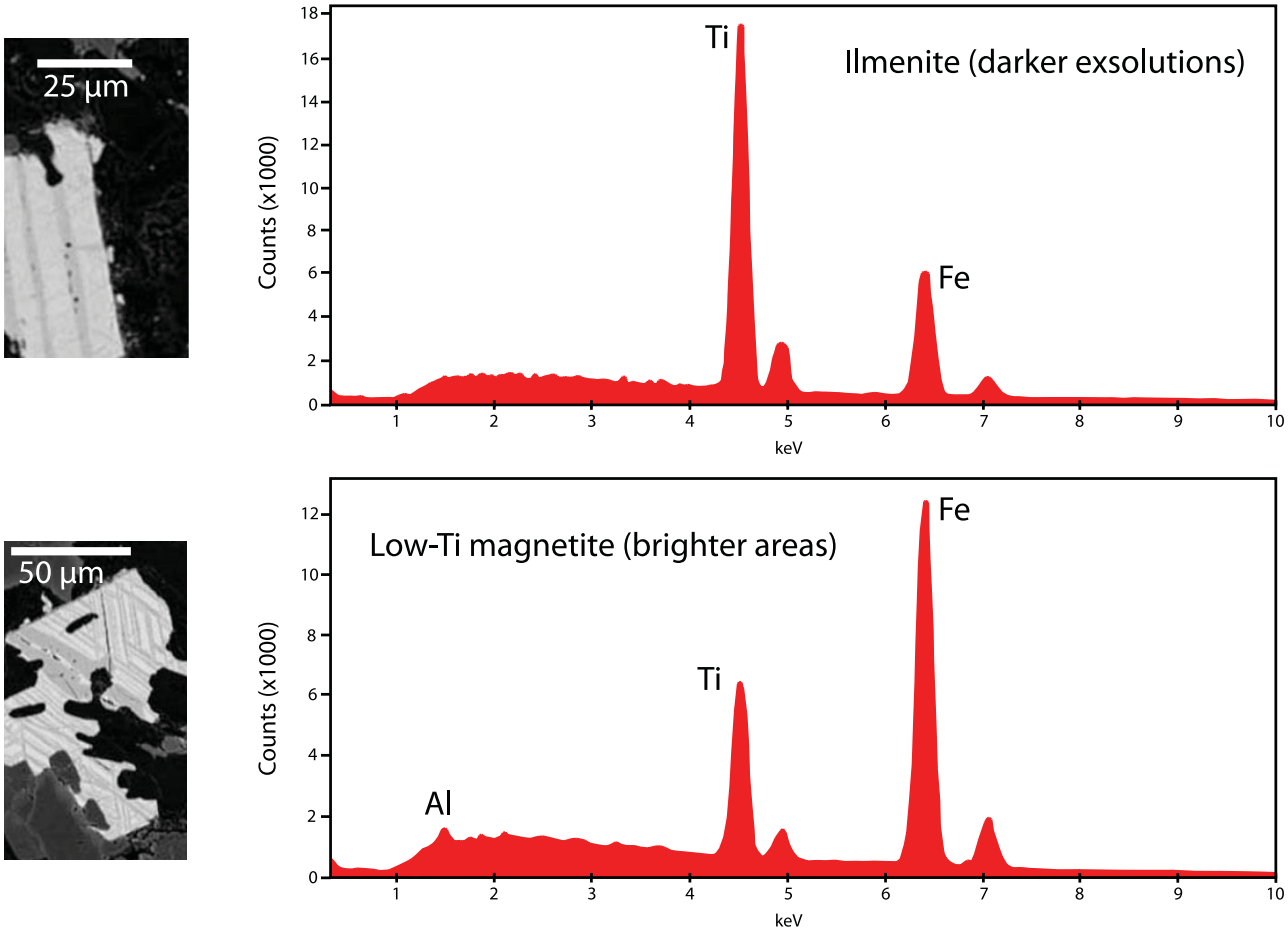



**Figure 4.** Scanning electron microscope images and energy dispersive spectra of Ti–Fe oxides with exsolution lamellae in sample CR379 (site 64).

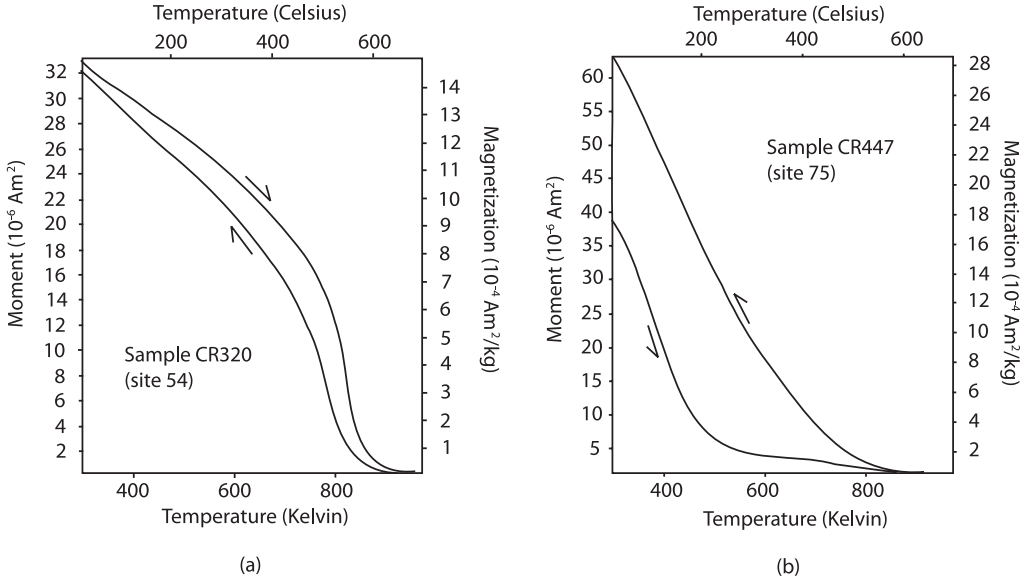



**Figure 5.** Magnetic susceptibility moments as a function of temperature. (a) Sample CR320 (site 54) reveals the presence of magnetite. The heating and cooling curves are nearly reversible, suggesting that the only magnetic component in this sample is low-Ti magnetite. (b) Sample CR447 (site 75) indicates the presence of a low-temperature component before 200 °C (likely titanomagnetite). The irreversibility of the heating and cooling curves suggests that the low-temperature component is metastable and easily altered to a higher temperature magnetic mineral, possibly magnetite.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

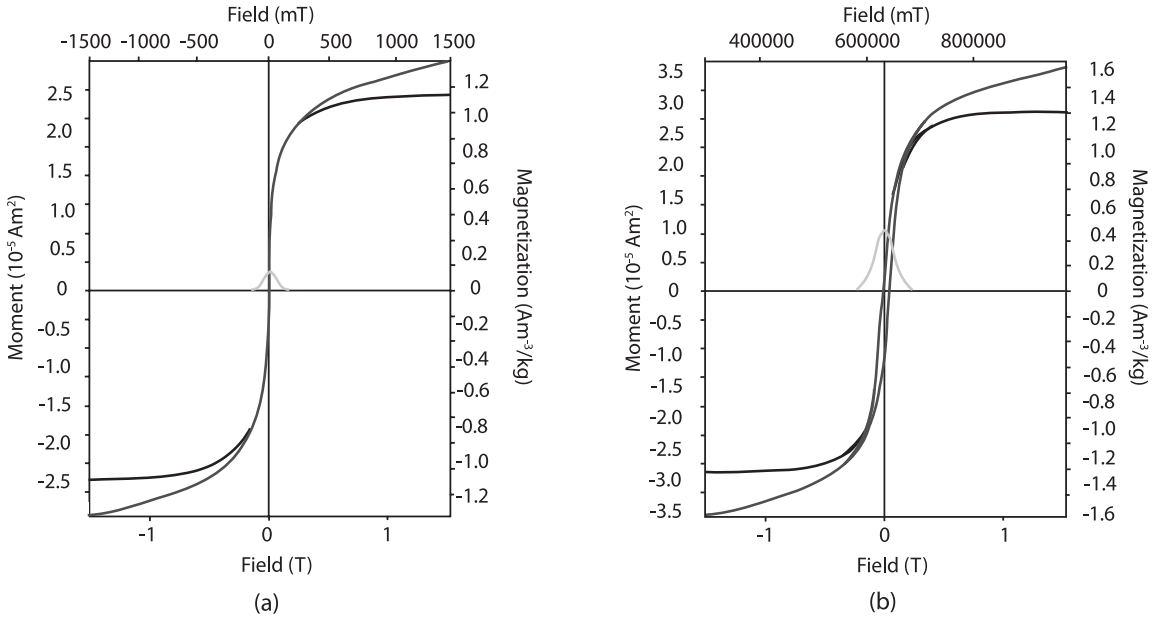



**Figure 6.** Hysteresis analysis showing (a) the hysteresis loop for sample CR19 (site 4) at 300 K, showing multidomain behaviour, and that of (b) sample CR320 (site 54) at 300 K, indicating PSD behaviour.

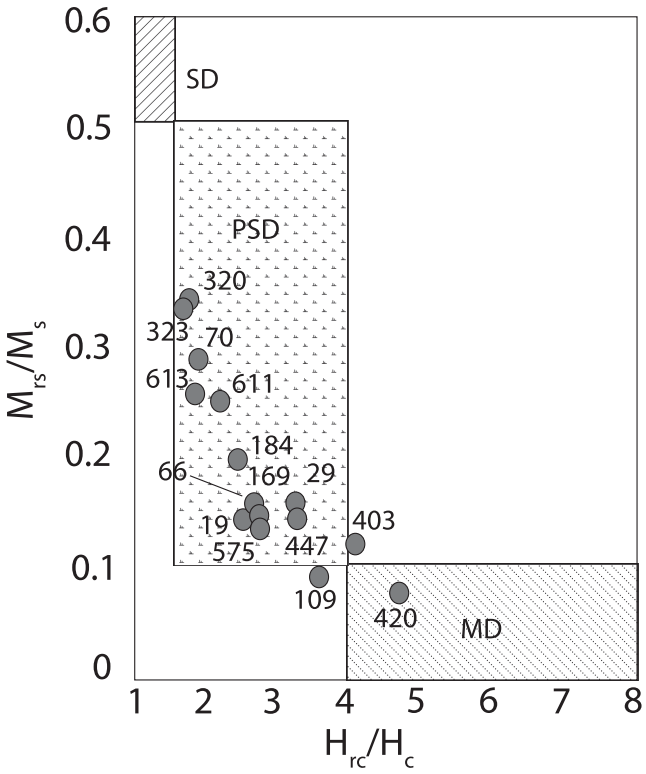



**Figure 7.** Day plot (Day *et al.* 1977), revealing that the majority of the samples for this study contain PSD grains. SD = single domain and MD = multidomain.

more. Five of these sites, however, seem unusable for a tilt test, because they have transitional or anomalous site-mean directions in tilt-corrected and *in situ* coordinates as well as anywhere in between. The remaining 10 sites have site means that converge upon tilt correction towards the overall formation-mean directions obtained from the large number of flat lying sites (Table 2; Fig. 10). Two sites are included in this group after inversion of R-polarity directions (∗ in Table 2). The 10 site means of these tilted lavas pass the Tauxe & Watson (1994) test, albeit with large error margins (104 ± 45 per cent unfolding; see Fig. 10.

*Reversal test*. The mean of all 40 N-polarity site means, which pass the filters of grouping (see below) and Vandamme (1994) cut-off (Table 3) is dec/inc = 6.6°/+61.2° with $k = 29.3$ and $\alpha_{95} = 4.2°$, which can be compared with the overall mean of 44 R-polarity directions that are similarly filtered, and has dec/inc = 178.2°/−59.2°, $k = 16.0$ and $\alpha_{95} = 5.5°$ (Table 3). This comparison shows that the N and R polarity means are 175.4° apart. The reversal test is positive with a 'B' classification according to McFadden & McElhinny (1990).

*Age dating and magnetostratigraphy*. The CRB are well-dated as mid-to-late Miocene on the basis of a stratigraphy built upon K–Ar and $^{40}$Ar–$^{39}$Ar dating methodologies (Waters 1961; Watkins & Baksi 1974; McKee *et al.* 1977, 1981; Swanson *et al.* 1979a; Long *et al.* 1983; Tolan *et al.* 2009; Barry *et al.* 2010). Ages are listed and referenced in Table S4 and are entered on the side of a magnetostratigraphic column (Fig. 11) that is based on treating each flow as a unit, with non-sequential site numbers in stratigraphical order, as done in Table 1. This means that site 15 is the youngest flow (rank = 1, on top), whereas the oldest flow sampled in this study is site 111 (rank = 84). The figure shows declination, inclination and the N- and R-polarity determinations from our study (Table 1).

The youngest two normal polarities are obtained from basalts with ages between 6.0 and 8.5 Ma. These ages have been assigned in Fig. 11 to levels in our stratigraphic column by correlation of the reversal pattern for the Saddle Mountain basalts of Reidel *et al.* (2003), reproduced here as column d. The pattern of our flows below these ages is essentially R–N–R, with ages between 10.5 and 15.6 Ma, again assigned by polarity correlation. The patterns of declination and inclination are reasonably well defined throughout the columns; an exception exists at the boundary between the Wanapum and Saddle Mountain formations where site 24 shows an anomalous steeply down direction in an otherwise long *R* sequence.

Previously, K–Ar ages established the general sequence from the base of the mid-Miocene (15.6 Ma) to the late Miocene (6 Ma), and

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

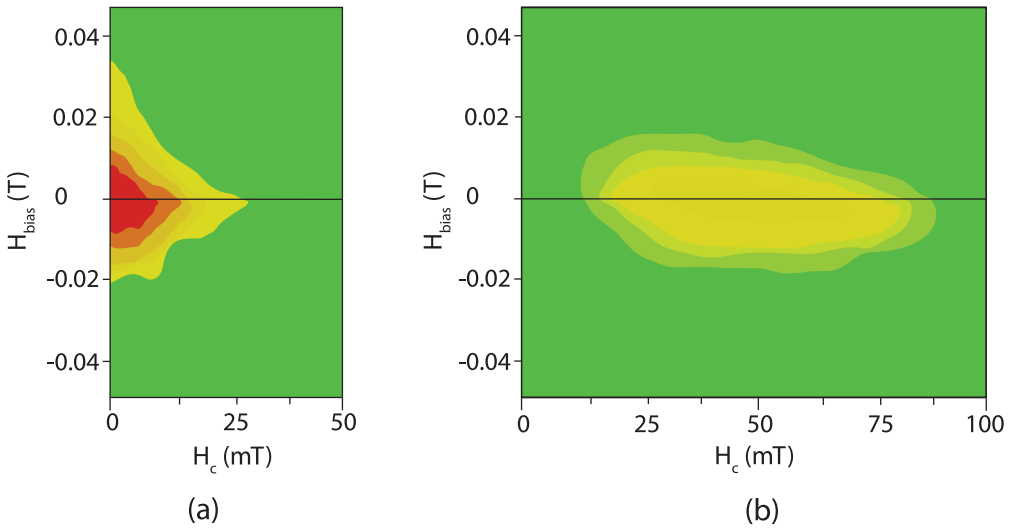



**Figure 8.** First-order reversal curves (FORC) for (a) sample CR19 (site 4) and (b) sample CR320 (site 54) at 300 K. Sample CR19 shows PSD behaviour, whereas sample CR320 shows single-domain behaviour.

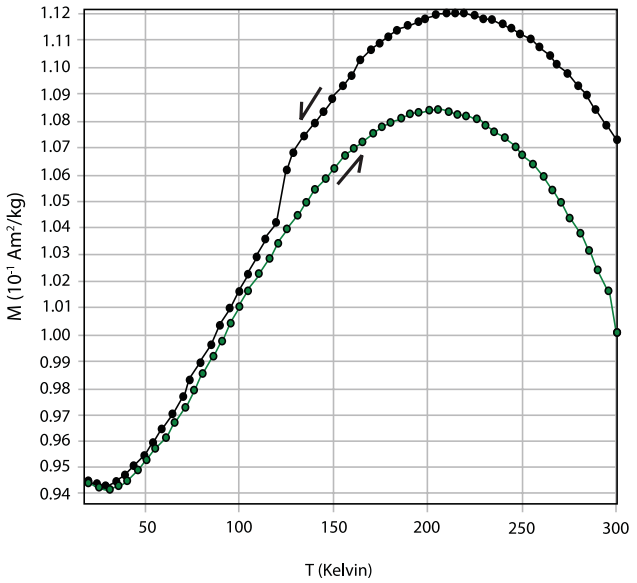



**Figure 9.** Remanent magnetization intensity upon cooling and then reheating (see arrows) of sample CR630 (site 117), showing the Verwey transition of magnetite at about 125 K.

have in the case of the Grande Ronde Formation been confirmed by $^{40}Ar/^{39}Ar$ ages (Barry *et al.* 2010). Details and references of the K–Ar and $^{40}Ar/^{39}Ar$ ages are listed in Table S4.

*Grouping*. For the determination of PSV it is common practice to use the mean of those site-mean directions that are similar in successive flows, based on the reasoning that such repeated site-mean directions may belong to flows that were erupted in fast succession and should count only as a single independent spot-reading of the geomagnetic field (i.e. as a VGP position). Before grouping the similar site means, the analysis starts by calculating the non-random ordering (NRO) factor (Biggin *et al.* 2008a) for each polarity interval that included five or more flows. A NRO factor of 0.95 and higher indicates that the data are not randomly distributed with 95 per cent confidence and can therefore be combined. The NRO factors and the indication to combine are included in Table 1 on the far right.

**Table 2.** Lava flows with significant bedding dip, used for the Tauxe & Watson (1994) tilt test.

| Rank | Site | dipdir/dip | Dec in-situ | Inc *in situ* | $k$ | $\alpha_{95}$ |
|---|---|---|---|---|---|---|
| 8 | 41* | 330/19 | 62.2 | 73.4 | 202.3 | 5.4 |
| 9 | 74 | 308/67 | 81.2 | 11.1 | 31.0 | 12.2 |
| 30 | 77* | 308/67 | 26.6 | 45.9 | 65.3 | 7.5 |
| 38 | 54 | 176/23.5 | 355.4 | 19.3 | 163.8 | 5.2 |
| 41, Gp4 | 26 | 342/15 | 155.3 | 88.0 | 201.7 | 6.5 |
| 51 | 55 | 181/17 | 350.3 | 33.0 | 517.4 | 4.0 |
| 56 | 59 | 188/33 | 10.1 | − 5.5 | 254.8 | 4.2 |
| 58, Gp9 | 44 | 202/36 | 1.1 | 7.9 | 912.4 | 2.2 |
| 58, Gp9 | 56 | 174/10.5 | 356.3 | 32.5 | 69.3 | 9.3 |
| 58, Gp9 | 105 | 185/19 | 359.9 | 26.2 | 37.8 | 12.6 |

Rank corresponds to the stratigraphic ordering (see Table 1). Gp denotes grouped site means (see Table 1). Asterisks denote sites with reversed-polarity directions that have been inverted to normal polarity. dipdir is dip direction.
Sites 68, 76, 78 and 115 have not been included because they have anomalous directions, discarded from the interpretation. Site 75 has been omitted because of its poor precision ($k = 9.7$).

Grouping only proceeded when the directions were within each other's cone of 95 per cent confidence ($\alpha_{95}$); this was the case for 33 sites. In Table 1, the resulting, grouped entries are labelled 'Gp x' ($x = 1, 2, 3.....13$) and are followed by the constituent sites. When we analyse mean directions or VGPs later on, the calculations can therefore be done with or without the grouped directions. Fig. 12 shows grouped and ungrouped site means for the collection; it can be seen that the difference is notable but not very large. The available radiometric ages can be satisfactorily assigned to our column of reversal boundaries (Fig. 11), but are not identifiable for individual flows that we sampled.

*Quality filtering and rejected sites*. McElhinny & McFadden (1997) list eight criteria that should be met, as a minimum, for PSV dispersion analysis. These eight criteria are reproduced here as Table S5. All have been met by our study, and the same is true for the reliability criteria of McElhinny & McFadden (2000, pp. 228–232). When a given site mean did not meet the desired threshold of values (see footnotes to Table S5), this result was excluded from the dispersion analysis. Rejected sites have also been marked in Table 1, using asterisks and italics.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

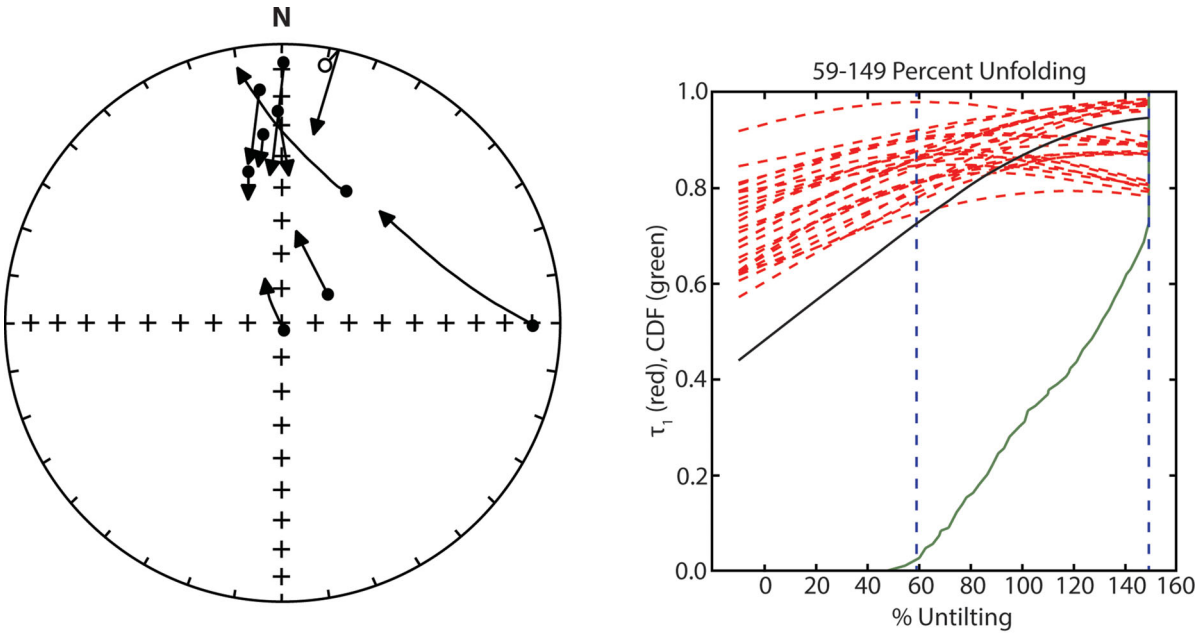



**Figure 10.** (a) Directions before (filled and open circles) and after (arrowheads) 100 per cent untilting (see Table 2 and text for details. (b) Tauxe & Watson (1994) positive tilt test. The vertical blue dashed lines enclose 95 per cent of the maximum $\tau 1$ values, and constitute the error margins. The red lines are examples of the 500 bootstraps of the analysis, whereas the black line is the average of all 500 bootstraps. The green line is the cumulative distribution of the maximum eigenvalues ($\tau 1$) of each bootstrap.

**Table 3.** Overall Columbia River Basalt Group means, sorted by polarity, cut-off-angle, grouping and N2 $\geq$ 4.

| | Normal polarity | | | | | Reversed polarity | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Dec. | Inc. | N | $k$ | $\alpha_{95}$ | Dec. | Inc. | N | $k$ | $\alpha_{95}$ |
| Grouped, 45-degree cut-off | 6.6 | 61.2 | 40 | 29.3 | 4.2 | 178.2 | −59.2 | 44 | 16.0 | 5.5 |
| Grouped, Van Damme cut-off | 6.6 | 61.2 | 40 | 29.3 | 4.2 | 178.2 | −59.2 | 44 | 16.0 | 5.5 |
| Grouped, Vandamme cut-off, N2 $\geq$ 4 | 5.4 | 61.2 | 39 | 31.4 | 4.2 | 179.2 | −59.6 | 41 | 15.8 | 5.8 |
| Ungrouped, 45-degree cut-off | 6.0 | 60.7 | 56 | 30.4 | 3.5 | 179.7 | −60.2 | 48 | 16.8 | 5.2 |
| Ungrouped, Van Damme cut-off | 6.0 | 60.7 | 56 | 30.4 | 3.5 | 179.7 | −60.2 | 48 | 16.8 | 5.2 |
| Ungrouped, Van Damme cut-off, N2 $\geq$ 4 | 4.8 | 60.4 | 48 | 31.1 | 3.8 | 180.3 | −60.2 | 44 | 16.5 | 5.5 |
| Mean of all options | 5.9 N-lat | 60.9 E-long | 6 | 27 667 | 0.4 | 179.2 N-lat | −59.8 E-long | 6 | 15 066 | 0.5 |
| Palaeopole, based on overall polarity means | 84.4 | 69.0 | | | | 84.0 | 15.8 | | | |

Palaeopoles have been calculated for a common locality at 46.2 N, 117.3 W.

Very recently, the criteria have been made more strict in terms of the within-site precision parameter ($k_w$) and the number ($N_2$) of demagnetized samples of a given site (Cromwell *et al.* 2013). The number $N_2$ is generally equal to or greater than 5; exceptions are found for sites with $N_2 = 3$ that are included in group means and hence, are still contributing to a well-determined entry that is used for the dispersion calculation as well as meeting the threshold. In Table 3, means of site means are calculated for N- and R-polarity sites. Two calculations are entered in Table 3 with the condition $N_2 \geq 4$. It can be seen that these two entries are not significantly different in direction nor in a statistical sense from calculations that include sites with lower $N_2$. The condition that $k_w$ must be greater than 50 is frequently not met. However, only in a couple of exceptions is $k_w$ less than 20, as can be seen in Table 1.

As already mentioned, sites 85 and 97 did not yield coherent results and are excluded. All other site means, including those listed in Table 4 do meet the eight criteria of McElhinny and colleagues. As Fisher (1953) statistics do not have meaning for $N_2 = 2$, it has been replaced by the condition that $k_w$ for the corresponding site must be greater than 80.

It must be noted that site numbers were logically assigned in the field in the order the sites were sampled. Stratigraphic positions, however, are not reflected by these site numbers, but are shown instead as 'Rank' in column 1 of Table 1. Rank and site numbers can be compared in Table 1, as well as in Tables S1 and S2 in the Supporting Information data set.

## Palaeomagnetic site means: patters and dispersion calculations

*Cut-offs.* A small set of tables documents the choices made in preparation for the dispersion analysis. Table 1 lists the palaeomagnetic site means, sampling site (GPS) coordinates, VGPs and statistical parameters, in stratigraphic (or rank) order, and includes groupings (Gp 1–13). Listed in Table 1 with a double asterisk are sites that have mean directions that are located more than some

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

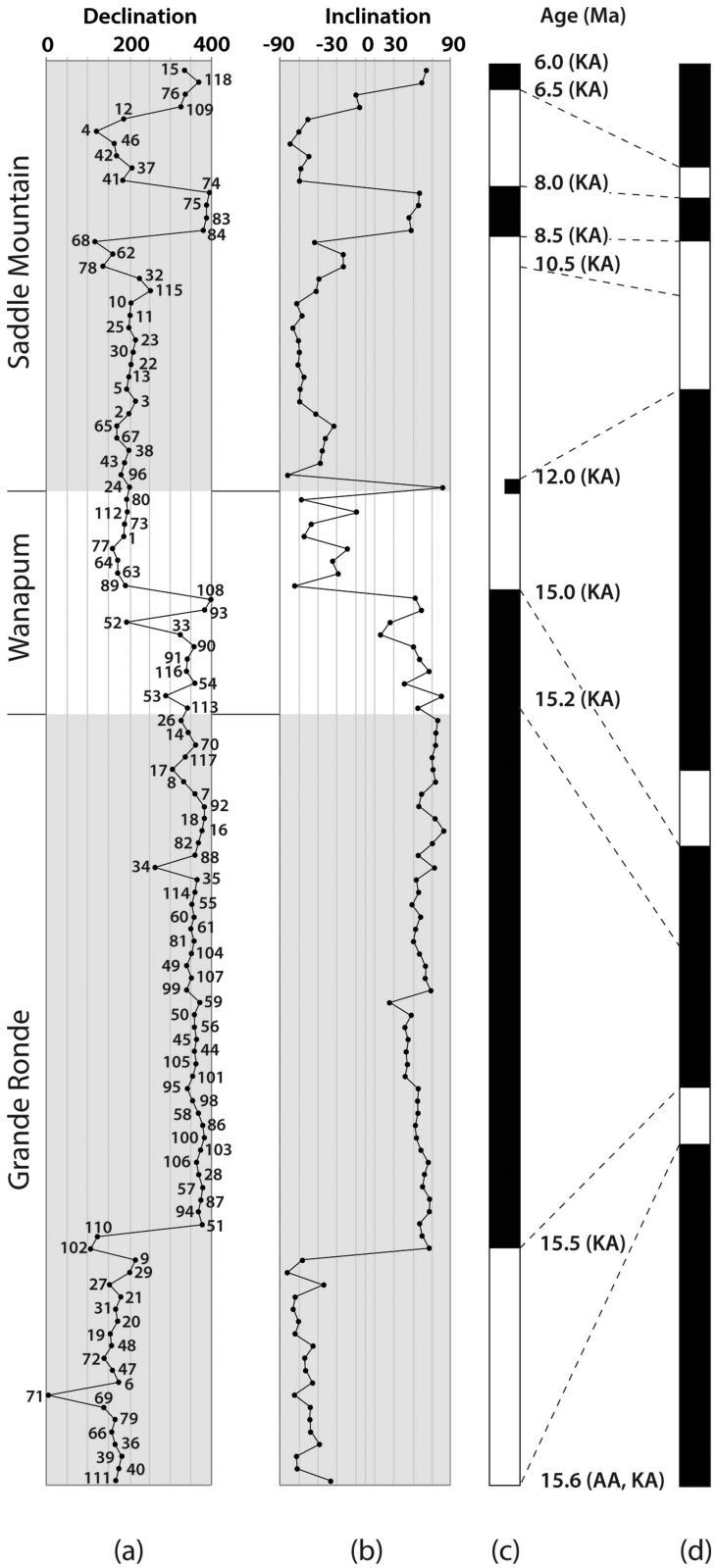



**Figure 11.** Magnetostratigraphy of the sampled flows for this study showing (a) its declination with corresponding site numbers, and (b) inclination plotted in stratigraphic order of the sites. Site 24 had a steeply down and southwestern direction and was diagnosed as an outlier; its N polarity is marked as a half-column, indicating its lack of reliability. The polarity of our collection (c) was then compared to the magnetostratigraphic columns from (d) Reidel *et al.* (2003). The (a) an (b) columns are alternatingly shaded according to the corresponding formation. Symbols AA or KA indicate that the corresponding ages were determined using $^{40}Ar/^{39}Ar$ or K–Ar methods, respectively. Ages with K–Ar or $^{40}Ar/^{39}Ar$ methodology are detailed in Table S4.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

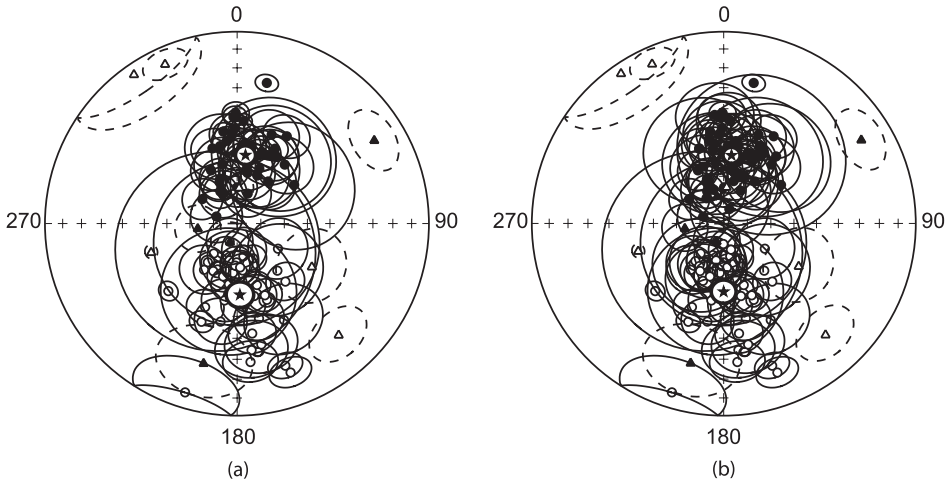



**Figure 12.** Site means with directions (a) grouped with Vandamme (1994) cut-off (45°), and (b) ungrouped with Vandamme cut-off (43.96–45.16°). Both normal and reverse polarities were recorded in this study; overall means are shown as stars. Outliers are depicted as closed or open triangles, with their corresponding cones of 95 per cent confidence ($\alpha_{95}$) as dashed ovals.

threshold value away from the overall mean. Dominguez (2012) explored three types of threshold: (1) one based on visual inspection, (2) a second one based on a fixed value of 45° and (3) one that follows the method of Vandamme (1994). The latter gave a threshold value of 45.16° for grouped directions and 43.96° for ungrouped. All three threshold types resulted in identifying similar outliers. To save space, we have only retained the Vandamme cut-off (rejecting entries in Table 1 in italics and marked by two asterisks). Readers interested in comparing the other cut-off outcomes are referred to the thesis of Dominguez (2012), which is publicly accessible. Fig. 12 shows all site-mean directions, grouped (in a) and ungrouped (in b), with eliminated site means displayed as triangles and dashed ovals of 95 per cent confidence. Table 4 includes the means for the grouped entries after Vandamme cut-off, sorted by polarity.

*Identifying patterns*. Linear regressions were carried out on the stratigraphically ordered directions, both on declinations and on inclinations. Site-mean directions beyond the Vandamme cut-off and low-precision sites were eliminated, and reversed directions were inverted, before the analysis. The regression of declination values was carried out, in order to determine whether there is a systematic trend as a function of age, such as would be observed if rotations affected the sampling region. A best fit of only 0.11° per million years is observed ($R = 0.0529$), indicating that negligible Miocene rotation influences the secular variation analysis. A similar regression was carried out on the inclinations and yielded 0.38° per million years (albeit also with a low $R = 0.0831$), indicating—when combined with the lack of declination trend—that apparent polar wander is not likely to be a significantly contributing trend in the palaeomagnetic directions from the 15.6 to 6.5 Ma interval.

An analysis of the magnitude of the elongation of the site-mean collections was performed on N + R directions by Dominguez (2012, her table 21) and yielded a mean value of 1.9 $_{-0.5}\backslash^{+1.0}$. The corresponding north–south elongation (strike of 356°) can be discerned in Fig. 12, especially in the reversed group of directions. The mean value of 1.9 overlaps, within error, with the elongation corresponding to an inclination of ∼60° in the TK03 model (Dominguez 2012; see Fig. 15e). The VGP distribution has similarly been analysed and yields an elongation factor of only 1.04 $_{-0}\backslash^{+0.75}$, indicating near-perfect circularity. We will return to the implications of these outcomes in the next section, when discussing the data set to be analysed in terms of dispersion.

*To analyse the dispersion of VGPs or that of directions?* Dispersion ($S$) quantifies the 'spread' of the entries (directions, VGPs) about their mean. Having grouped and applied appropriate data selection as discussed above, the choice between directions and VGPs depends on the circularity or ellipticity of their distribution. Published PSV analyses have generally proceeded on the assumption that the VGP distribution should be circular (see Tauxe 2010, p. 217), although exceptions to this have been suggested (e.g. Kono 1997), on the basis of results from the last 5 million years. Circularity of the VGP distributions implies that the collection of corresponding site-mean directions for mid-range palaeolatitudes is, or should be, meridianally elongated. Our directions are elongated indeed, whereas the VGPs are circularly distributed about their mean. Figs 15(a)–(d) is self-explanatory, and Fig. 15(e) plots the two elongation values allowing a comparison with the E/I (= elongation/inclination or elongation/latitude) curve of TK03 (Tauxe & Kent 2004). The clear contrast between the two elongation factors validates our choice of VGPs as the set to be analysed for dispersion.

The circularity of the VGP distribution implies that the corresponding site-mean directions cannot be circularly distributed. This is indeed what we observe (Fig. 15). McFadden *et al.* (1988, 1991), in discussing model G, argued that $S$ is a combination of two relatively independent sets of geomagnetic field families, one being symmetric about the equator ($S_Q$, quadrupolar, hexadecapolar and higher) and the other being antisymmetric ($S_D$, dipolar, octupolar and higher). Thus,

$$S(\lambda) = \sqrt{S_D^2 + S_Q^2} = \sqrt{(a\lambda)^2 + b^2},$$

where λ is palaeolatitude and $a$ and $b$ are constants. The $S_Q$ family dominates at the equator, where $S(\lambda) = S_Q = b$.

A database consisting of 3719 lava flows and thin dikes with ages 0–5 Ma was analysed in terms of latitude variation of the angular dispersion of the VGPs and compared with model G (McElhinny & McFadden 1997). The authors cautioned that careful evaluation of the efficacy of employed demagnetization methods is required

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

**Table 4.** Dispersion calculation for N or R polarity, grouped, with Vandamme (1994) cut-off.

| Normal polarity | | | | | | | Reversed polarity | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Site/GP | Site plat | *k* | *K* | *n* | Delta | Sw/sqt (n) | Site/GP | Site plat | *k* | *K* | *n* | Delta | Sw/sqt (n) |
| GP3 | 47.027 | 50.1 | 23.3 | 8 | 22.1442 | 5.9352 | GP1 | −42.691 | 44.2 | 23.3 | 10 | 18.6662 | 5.3067 |
| GP4 | 46.495 | 24.2 | 11.4 | 7 | 14.4264 | 9.0614 | GP2 | −42.618 | 475.5 | 251.2 | 4 | 17.2076 | 2.5554 |
| GP5 | 46.661 | 156.0 | 73.2 | 3 | 20.1781 | 5.4645 | 1 | −42.927 | 67.4 | 35.3 | 7 | 8.5678 | 5.1547 |
| GP6 | 46.485 | 60.5 | 28.5 | 12 | 10.9567 | 4.3765 | 2 | −42.936 | 78.4 | 41.0 | 6 | 19.1051 | 5.1631 |
| GP7 | 47.009 | 389.5 | 181.1 | 3 | 14.4071 | 3.4752 | 3 | −42.937 | 43.5 | 22.8 | 5 | 25.7611 | 7.5931 |
| GP8 | 46.984 | 17.7 | 8.2 | 6 | 11.9556 | 11.5234 | 4 | −42.666 | 23.1 | 12.2 | 7 | 39.8803 | 8.7705 |
| GP9 | 46.921 | 844.5 | 393.6 | 5 | 21.4665 | 1.8259 | 6 | −42.662 | 55.9 | 29.5 | 6 | 8.0743 | 6.0894 |
| GP10 | 47.090 | 44.4 | 20.6 | 7 | 16.2363 | 6.7459 | 9 | −42.695 | 42.4 | 22.3 | 6 | 24.7396 | 6.9953 |
| GP11 | 47.059 | 25.8 | 12.0 | 9 | 21.7620 | 7.8013 | 12 | −42.686 | 81.8 | 43.1 | 2 | 4.9998 | 8.7221 |
| GP12 | 46.757 | 38.2 | 17.9 | 13 | 5.0231 | 5.3119 | 19 | −42.597 | 143.5 | 75.9 | 6 | 23.3635 | 3.7969 |
| GP13 | 46.964 | 25.8 | 12.0 | 14 | 13.3175 | 6.2466 | 20 | −42.594 | 170.3 | 90.0 | 4 | 6.6934 | 4.2684 |
| 7 | 46.548 | 79.9 | 37.6 | 6 | 6.5681 | 5.3905 | 21 | −42.591 | 98.2 | 51.9 | 6 | 12.4364 | 4.5894 |
| 8 | 46.534 | 89.2 | 42.0 | 6 | 22.9325 | 5.1008 | 23 | −42.585 | 295.9 | 156.5 | 6 | 26.4497 | 2.6436 |
| 15 | 46.521 | 62.7 | 29.6 | 6 | 19.9879 | 6.0828 | 25 | −42.585 | 75.5 | 39.9 | 6 | 23.3138 | 5.2336 |
| 17 | 46.493 | 61.2 | 28.9 | 6 | 36.2172 | 6.1545 | 27 | −42.248 | 24.4 | 13.0 | 5 | 27.7463 | 10.034 |
| 50 | 46.927 | 460.2 | 214.5 | 5 | 17.5166 | 2.4736 | 29 | −42.585 | 70.1 | 37.1 | 6 | 33.1289 | 5.4314 |
| 51 | 46.927 | 28.3 | 13.2 | 6 | 17.2367 | 9.1059 | 31 | −42.592 | 282.0 | 149.1 | 5 | 16.3862 | 2.9668 |
| 53 | 46.928 | 37.8 | 17.6 | 6 | 41.2158 | 7.8791 | 32 | −42.608 | 265.7 | 140.4 | 6 | 37.9539 | 2.7908 |
| 54 | 46.945 | 163.8 | 76.3 | 6 | 23.0011 | 3.7859 | 36 | −43.030 | 45.5 | 23.7 | 6 | 17.9079 | 6.787 |
| 55 | 46.929 | 517.4 | 241.1 | 4 | 18.5701 | 2.6083 | 37 | −43.032 | 66.3 | 34.6 | 6 | 19.0582 | 5.6226 |
| 58 | 46.948 | 82.5 | 38.4 | 6 | 11.5064 | 5.3348 | 38 | −43.036 | 199.3 | 104.0 | 6 | 22.7978 | 3.2431 |
| 59 | 46.948 | 254.8 | 118.7 | 6 | 34.0343 | 3.0356 | 39 | −43.040 | 153.6 | 80.1 | 5 | 10.3583 | 4.0471 |
| 60 | 46.948 | 120.6 | 56.2 | 6 | 7.5092 | 4.4124 | 40 | −43.041 | 87.4 | 45.6 | 6 | 8.2151 | 4.8977 |
| 70 | 47.112 | 64.1 | 29.7 | 6 | 14.1202 | 6.0662 | 41 | −43.046 | 202.3 | 105.5 | 5 | 9.7362 | 3.5267 |
| 74 | 46.921 | 31.0 | 14.4 | 6 | 30.6063 | 8.6996 | 42 | −43.042 | 96.4 | 50.3 | 5 | 8.4091 | 5.1087 |
| 83 | 46.986 | 41.2 | 19.2 | 7 | 29.7136 | 6.9929 | 43 | −43.036 | 78.8 | 41.1 | 4 | 16.3338 | 6.3169 |
| 84 | 46.984 | 36.7 | 17.1 | 6 | 23.0484 | 8.0026 | 46 | −43.026 | 67.1 | 35.0 | 5 | 26.4158 | 6.1218 |
| 88 | 46.994 | 105.3 | 49.0 | 5 | 9.9083 | 5.1761 | 47 | −43.051 | 177.3 | 92.4 | 5 | 13.5745 | 3.7675 |
| 90 | 47.013 | 36.9 | 17.2 | 8 | 15.7542 | 6.9144 | 48 | −43.053 | 475.4 | 247.9 | 6 | 18.7436 | 2.1004 |
| 91 | 47.028 | 65.7 | 30.5 | 7 | 14.7846 | 5.5408 | 62 | −43.045 | 115.1 | 60.0 | 6 | 35.4382 | 4.2681 |
| 92 | 47.032 | 69.6 | 32.3 | 4 | 19.9082 | 7.1219 | 63 | −43.066 | 58.5 | 30.5 | 5 | 32.7898 | 6.5603 |
| 94 | 47.058 | 45.3 | 21.0 | 5 | 6.7443 | 7.8987 | 64 | −43.065 | 24.9 | 13.0 | 6 | 29.2485 | 9.1793 |
| 99 | 47.123 | 76.2 | 35.3 | 5 | 16.4971 | 6.0957 | 65 | −43.073 | 1059.0 | 551.8 | 6 | 26.5970 | 1.4077 |
| 102 | 47.099 | 32.7 | 15.2 | 6 | 21.8607 | 8.4916 | 66 | −43.079 | 89.9 | 46.8 | 6 | 15.8899 | 4.8319 |
| 103 | 47.096 | 91.8 | 42.6 | 6 | 12.5534 | 5.0679 | 67 | −43.079 | 52.4 | 27.3 | 6 | 20.3249 | 6.329 |
| 110 | 47.010 | 89.0 | 41.4 | 2 | 37.3175 | 8.904 | 69 | −43.122 | 90.5 | 47.1 | 5 | 30.8574 | 5.2789 |
| 113 | 47.118 | 58.7 | 27.2 | 6 | 18.9841 | 6.3395 | 72 | −43.222 | 164.1 | 85.1 | 2 | 29.4099 | 6.2077 |
| 116 | 47.042 | 25.9 | 12.0 | 6 | 13.2718 | 9.5338 | 73 | −43.067 | 51.9 | 27.0 | 6 | 8.6533 | 6.3582 |
| 117 | 47.024 | 70.1 | 32.6 | 6 | 18.2442 | 5.7936 | 77 | −43.041 | 65.3 | 34.1 | 7 | 37.9371 | 5.2459 |
| 118 | 47.002 | 199.7 | 92.9 | 6 | 7.1794 | 3.4315 | 79 | −43.125 | 70.0 | 36.4 | 5 | 11.0449 | 6.0026 |
| | | | | | | | 80 | −43.254 | 85.8 | 44.5 | 7 | 12.7736 | 4.5911 |
| | | | | | | | 89 | −43.133 | 56.6 | 29.4 | 4 | 18.5176 | 7.4643 |
| | | | | | | | 111 | −43.117 | 80.2 | 41.7 | 3 | 25.2578 | 7.2389 |
| | | | | | | | 112 | −43.089 | 19.7 | 10.3 | 5 | 41.2086 | 11.309 |

and that many of the older studies need to be redone using more modern methods. This suggestion provided the motivation for the present study, as well as for the TAFI (Johnson *et al.* 2008). We will return below to a concern about differences in $S$ for separate N- and R-polarity data sets (McElhinny & McFadden 1997).

In stark contrast with the reliance on $S = f(\lambda)$ as a characteristic of the dipole/octupole family geomagnetic fields, Linder & Gilder (2012) have argued that this latitude dependence of $S$ is a mathematical artifact and that it stems from the non-linear conversion of directions to VGPs. This would imply that the directions (dec, inc) should be Fisherian distributed and the VGPs should not. However, even in the case of non-compacted sedimentary or igneous rocks, the distributions often turn out to be elongated; the present study adds one more example from extrusives. Because most PSV analyses have been performed at the VGP-level, for comparison we have used a similar treatment.

*Dispersion parameters at VGP level for our sites*. Separate N- and R-polarity data sets, as listed in Table 4, have been analysed. The formula used is

$$S_b = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(\Delta_i^2 - \frac{S_{w_i}^2}{N_{s_i}}\right)}.$$

Here, $\Delta_i$ is the distance of the $i$th site's VGP to the overall mean of the VGPs, $N$ is the total number of sites, $S_w$ is the within-site

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

**Table 5.** Summary of the dispersion calculations.

| Data set | Polarity | Type | Dec. | Inc. | Mean palaeopole | Study mean | | *n* | *K* | $A_{95}$ | S(no_Ws) | $S_L$ | $S_b$ | $S_U$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | (mean) | (mean) | (lat, long) | Geogr coord | Palaeolat | | | | | | | |
| 1. This study | N | Gr[a] | 6.6 | 61.2 | 83.7N, 11.1E | 46.4N, 242.4E | 42.3 | 40 | 29.3 | 4.2 | 18.1 | 14.3 | 17.0 | 19.7 |
| 2. This study | R | Gr[a] | 178.2 | −59.2 | 83.5S, 254.6E | 46.3N, 242.7E | −40.0 | 44 | 16.0 | 5.5 | 23.6 | 20.1 | 22.9 | 25.5 |
| 3. This study | N | Ungr[a] | 6.0 | 60.7 | 83.6N, 17.7E | 46.4N, 242.4E | 41.7 | 56 | 30.4 | 3.5 | 16.9 | 13.6 | 15.6 | 17.5 |
| 4. This study | R | Ungr[a] | 179.7 | −60.2 | 84.4S, 245.2E | 46.3N, 242.7E | −41.1 | 48 | 16.8 | 5.2 | 24.7 | 20.9 | 23.9 | 26.7 |
| This study | N + R | Ungr[a] | 3.7 | 60.0 | 84.0N, 35.0E | 46.3N, 242.5E | 40.6 | 104 | 22.8 | 3.0 | 20.8 | 17.9 | 19.7 | 21.6 |
| 5. Published | N | Ungr[b] | 348.7 | 63.5 | 82.0N, 142.9E | 46.7N, 240.6E | 45.1 | 59 | 30.6 | 3.4 | 18.8 | 15.6 | 18.3 | 20.9 |
| 6. Published | R | Ungr[b] | 193.9 | −53.6 | 73.7S, 195.5E | 46.6N, 240.6E | −34.1 | 69 | 23.0 | 3.6 | 20.8 | 15.7 | 20.5 | 24.9 |

The numbers in column 'data set' correspond to those of Fig. 13. N and R indicate geomagnetic polarity.
Type: Gr, grouped; Ungr, ungrouped.
[a]Vandamme cut-off applied.
[b]The previously published data (see text) are in approximate stratigraphic order, but without knowledge about superposition; therefore, these data cannot be grouped. No cut-off has been applied to these data, which are already filtered in the original.
Palaeolat = palaeolatitude of the mean locality as calculated from the mean palaeopole.
*n* = number of entries, *K* and $A_{95}$ are the statistical parameters (Fisher 1953), as calculated at the formation level.
*S*(no_Ws) is a factor that accounts for within-site scatter of directions. $S_b$ is the between-site angular dispersion.
$S_L$ and $S_U$ are the 95 per cent lower and upper bounds on the value of $S_b$, using 10 000 bootstraps.

**Table 6.** Modelling of dispersion data with model G, compared with our new data.

| | a | a-high | a-low | b | b-high | b-low | λ | $S_{bm} = \sqrt{(a^2+b^2\lambda^2)}$ | $S_{bm}$-highest | $S_{bm}$-lowest | $S_b$ | $S_b$-high | $S_b$-low |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| TAFI | 14.6 | 15.6 | 13.4 | 0.20 | 0.24 | 0.13 | 46.35 | 17.29 | 19.16 | 14.69 | | | |
| CRB-N | | | | | | | 46.40 | | | | 17.00 | 19.70 | 14.30 |
| CRB-R | | | | | | | 46.30 | | | | 22.90 | 25.50 | 20.10 |
| CNS-1 | 8.7 | 10.7 | 6.3 | 0.27 | 0.31 | 0.22 | 46.35 | 15.24 | 17.91 | 11.99 | | | |
| CNS-2 | 9.9 | 11.3 | 7.8 | 0.26 | 0.29 | 0.22 | 46.35 | 15.60 | 17.56 | 12.84 | | | |
| Jur-1 | 16.4 | 19.0 | 10.3 | 0.19 | 0.46 | 0.00 | 46.35 | 18.61 | 28.56 | 10.30 | | | |
| Jur-2 | 10.8 | 12.0 | 8.3 | 0.24 | 0.32 | 0.14 | 46.35 | 15.50 | 19.08 | 10.54 | | | |

Dispersion ($S_b$) and its error limits (S-high, S-low) as fitted with model G (subscript m, $S_{bm} = \sqrt{(a^2+b^2\lambda^2)}$) of published data sets, compared with the dispersion parameters from this study (CRB), shown in the three columns on the far-right side of the table. These results are detailed in Table 5, numbers 1 and 2 (grouped, Vandamme cut-off applied, N and R are normal and reversed polarity data sets, respectively).
TAFI consists of results 0–5 Ma, from Johnson *et al.* (2008), as summarized by Biggin *et al.* (2008a, Fig. 9); CNS (Biggin *et al.* 2008a) contains data from the Cretaceous.
Normal Superchron, and Jur contains data sets from the Jurassic (Biggin *et al.* 2008a). Sets 1 and 2 represent different quality-filtered data, labelled group 1 and group 2, respectively.

dispersion and $N_s$ is the number of samples in the *i*th site (see eq. 4 in Johnson *et al.* 2008). $S_w$ is approximated by the square root of (81°)/*K*, where *K* is the precision parameter at the VGP level as approximated from *k* (using eq. 12 of Cox 1970).

The values of $S_b$ and its error limits can be seen in Table 5. Separate analyses have been performed for grouped and ungrouped data, and for N and R polarity. All analyses were performed after Vandamme cut-off was applied. Also included are the outcomes of dispersion calculations for published palaeomagnetic directions from the stable interior of the CRB, which we will discuss below. Fig. 13 visualizes the three *N* and the three *R* means listed in Table 5. The six $S_b$ values derived from our present study have been plotted in Fig. 14. The R-polarity results are significantly higher than any of the other values from the literature for this latitude span, whereas the N-polarity results are just slightly higher. The $S_b$ values from our collection are compared in Fig. 14 with TAFI data, a best-fit curve for 195–5 Ma, and Precambrian values from Smirnov *et al.* (2011). Despite some scatter and occasional overlap, the general impression is that our Miocene data-points reveal a higher dispersion (especially for our R-polarity results) than for the Precambrian data. The upper two best-fit lines in Fig. 14 are for the TAFI analysis (Johnson *et al.* 2008) and the Jurassic to Miocene results from McFadden *et al.*'s model G (1991), as reproduced in Smirnov *et al.* (2011). These trends pass through the N-polarity, but not the R-polarity Miocene results.

*The dispersion parameters of previously published results.* The palaeomagnetism of the CRB has been extensively studied, with sites in rotated as well as non-rotated areas. There is sufficient information to perform a dispersion analysis also on these results, insofar as no indications exist for local or regional rotations. From a compilation made by Drs Jonathan T. Hagstrum and Michael G. Sawlan, we have extracted the (published and otherwise suitable) results from the CRB in (mostly) south-central Washington. These are listed in Table S3. Stratigraphically these data cover the same three formations as our own sites: Grande Ronde, Wanapum and Saddle Mountain. The same filtering applies, in terms of Vandamme cut-off (43.96–45.16°), poor or unknown statistical parameters, exclusion of clearly rotated localities.

One process that could not be performed, however, is the grouping, because we lack knowledge about which flows (sites) are in contact with each other.

The $S_b$ parameters for this collection turn out, reassuringly, to be quite compatible with the results from our own sampling, as can be seen in Table 5 (results 5 and 6; see also Fig. 13). The observation, in our sampled sites, that N-polarity results have lower $S_b$ values

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

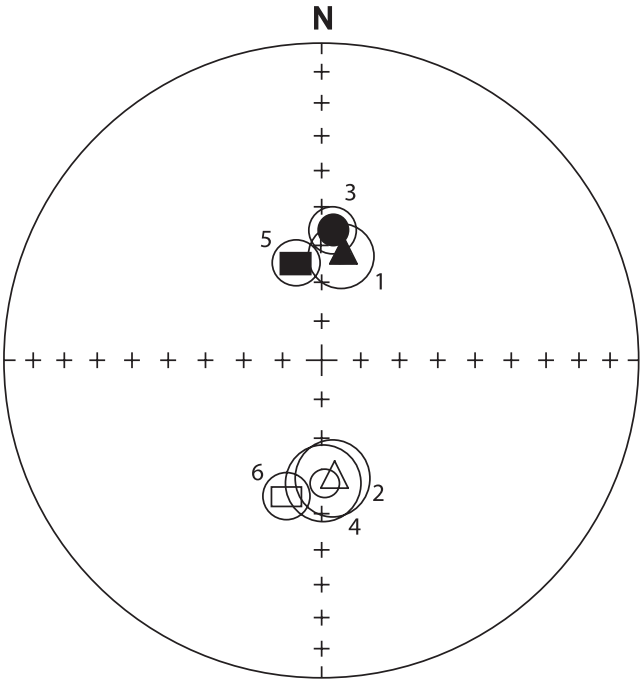



**Figure 13.** Equal-area stereonet with the means of the various grouping and cut-offs applied numbers correspond to those of Table 5: 1 = grouped, N polarity, Vandamme cut-off; 2 = grouped, R polarity, Vandamme cut-off; 3 = ungrouped, N polarity, Vandamme cut-off; 4 = ungrouped, R polarity, Vandamme cut-off; 5 = ungrouped, N polarity, previously published data; 6 = ungrouped, R polarity, previously published data).

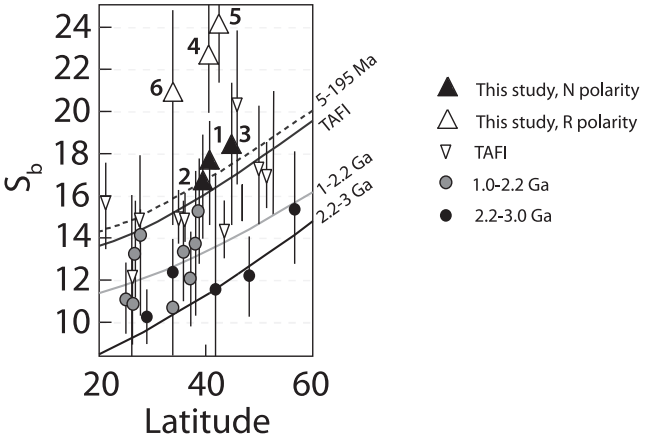



**Figure 14.** VGP scatter, $S_b$, as a function of latitude for the three N-polarity and three R-polarity site-mean populations of this study (upright triangles, with numbers corresponding to those of Fig. 13 and Table 5). Comparative values of $S_b$ are shown as four sloping lines for the same latitude interval, for four analyses: one is the time-averaged field initiative (TAFI; Johnson *et al.* 2008), a second one involves data between 5 and 195 Ma (*McFadden et al.*'s model G, as reproduced in Smirnov *et al.* 2011), and two more lines are for two Precambrian intervals (Smirnov *et al.* 2011) with ages of 1.1–2.2 and 2.2–3.0 Ga. Individual data point that fall inside the box have been retained, so that it can be seen that our data generally have higher dispersion, especially those of R polarity (open upright triangles), than the other results.

than the significantly higher values for the R-polarity results, is also seen in this collection of previously published data.

## DISCUSSION

Our study of 118 flows has yielded 84 geomagnetic field directions, after grouping was carried out to eliminate successive statistically similar directions, and after application of the Vandamme cut-off. The directions do not reveal trends, neither reflecting rotations, nor significant apparent polar wander.

The dispersion parameters calculated for various options in grouping, cut-offs and polarity choices are generally all on the high side, when compared with data from other analyses of Neogene and Jurassic sample collections, although the error limits overlap. When compared with results for the CNS, however, the latter appear to have significantly and robustly lower dispersion. We have carried out the analysis at the VGP level, given that the VGP distributions are clearly circular, whereas the palaeomagnetic directions are elongated along the meridian. These attributes are in contradiction to the recommendations of Linder & Gilder (2011, 2012), who have postulated circular direction distributions and, hence, advocate to analyse the direction distributions instead of VGP distributions. As an aside, the elongation characteristics of the direction distributions may well be a preferred model to such an extent that it could be taken as a reliability criterion.

The relatively lower dispersions ($s_b$) of rocks magnetized during the CNS, as compared to the Neogene and Jurassic dispersions, are taken to indicate a correlation between reversal rates (zero for 118–84 Ma) and $s_b$, supporting the conclusions of Biggin *et al.* (2008a).

Our study can form the kick-starter for a research program studying Miocene and Palaeogene secular variations, as well as a planned comparison with Permian dispersion values in studies of the PCRS that are underway (Menzo 2013; Bazhenov *et al.* 2014).

*Differences in S for separate N- and R-polarity data.* The scatter in VGPs as well as in direction space (Figs 12 and 15c, Tables 3 and 5) is greater for R-polarity results than for N-polarity data. This can be seen by examining, for instance, the values of the precision parameter $k$ (or $K$). Table 3 shows that $k$ is typically 16 for the R collection of site means and about 30 for N-polarity data. In Table 5 the same contrast in $K$ is associated with the means of the VGPs as well as clearly observed in the dispersion parameter $S_b$.

It is not straightforward to find explanations for such a difference, which is statistically significant for the N and R dispersion parameters (Table 5, compare the N and R grouped entries and their lower and upper 95 per cent confidence limits on $S_b$). *A priori* one could attempt to attribute the cause to either one pertaining to the magnetic mineralogy (rock or palaeomagnetic in nature) or one that results from geomagnetic field properties and core-mantle boundary conditions.

McElhinny & McFadden (1997; MM97) observed a similar difference, with R-results showing greater dispersion. They remarked that the laboratory treatment, in terms of directional analysis of demagnetization, may have been wanting in older publications, suggesting that a palaeomagnetic factor contributed to the R-scatter. A minor decrease in the magnitude of the difference is seen when the results are filtered by a 'demagnetization code (DC)' (their fig. 6) and could be taken as support for their interpretation.

The distribution of our site-mean directions may have been affected by an unrecognized viscous (present-day field, PDF) overprint, such as the one blamed by MM97. The elongated site-mean distribution of Fig. 15(c) is approximately north–south (strike = 356°) and best fit by a vertical plane. In or near that plane we should find the N-mean, R-mean and the PDF direction.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

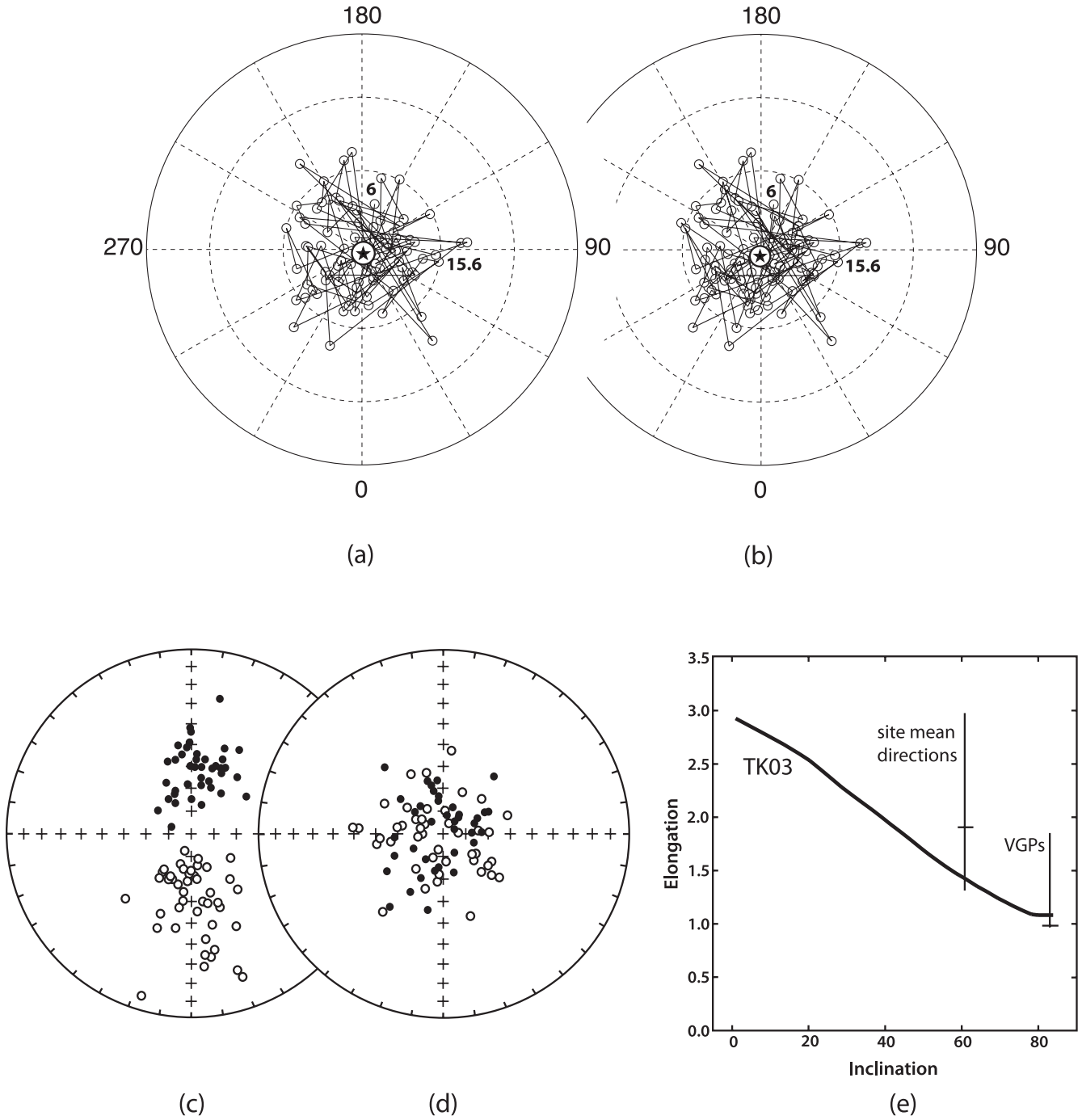



**Figure 15.** Mean poles (stars) and joined VGPs, indicative of secular variation for sites that were (a) grouped with Vandamme cut-offs and (b) ungrouped with Vandamme cut-offs. The numbers indicate the age of the youngest (6 Ma) and oldest (15.6 Ma) flows of the collection. (c) Site-mean directions and (d) VGPs for grouped entries filtered with Vandamme cut-off, and used in an E/I (elongation/inclination) analysis, as portrayed in (e) where it can be seen that the directions are streaking north-south, whereas the VGPs are nearly perfectly circularly distributed.

An unrecognized PDF will have an inclination (64°) that is slightly steeper (by 3.5°) than the Miocene N-mean which has an inclination of 60.3° (Table 5). Thus, it is possible that the effect of superposition of the PDF and N-mean causes a slight steepening of the resultant vector. By the same line of reasoning, an R-mean plus PDF would have a slightly shallower inclination, and indeed that is observed given the R-inclination as −59.3° (Table 5).

On the other hand, our demagnetization characteristics of N and R samples (Figs 3a, c, e, g, i) do not provide any evidence for an overprint; only occasionally, do we observe the curved trajectory (Fig. 3k) that could be taken as a hint of a PDF. At this time, convincing explanations for the difference in $S_b$ of N and R distributions of VGPs elude us, not in the least because the differences between the testable outcomes that we can examine are no larger than some

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

4°. Future tests should be designed to constrain the possible options better.

## CONCLUSIONS

Our collection of palaeomagnetic sites (flows) of the Miocene CRB has revealed three R and three N-polarity intervals. Radiometric age dating provides age brackets of some 9 million years (15.6–6.5 Ma) for the collection. Tilt and reversal tests are positive. Only two sites (out of 118) gave uninterpretable results, two other sites had poor precision ($k < 11$), and 10 sites yielded directions that deviated by more than 45° from the overall mean, thus being removed in a Vandamme cut-off procedure (1994). Sequential flows with mean directions falling within each other's cone of 95 per cent confidence have been grouped; this resulted in 13 groups representing 33 sites. The remaining 40 N-polarity and 44 R-polarity results have been analysed in terms of elongation of site-mean directions and VGPs and dispersal parameters of the VGP distribution. Previously published palaeomagnetic results from Columbia Plateau basalt sites in the same area as our sampling localities have also been analysed for dispersion.

The VGP population, with elongation factor $E = 1.04$, is circularly distributed, whereas the site-mean directions are elongated in a north–south meridianal plane ($E = 1.9$). Such patterns conform to the expectations that the VGPs should be the choice for determining the dispersion parameters. N-polarity results yield dispersion parameters ($S_b$) ranging from 15.6 to 18.3, whereas R-polarity results yield parameters ranging from 20.5 to 23.9, significantly higher than those of the N-sites. No indications are found in the demagnetization behaviour of an unrecognized viscous overprint that may have biased the site-mean directions; thus, this difference remains a challenge for future tests of this Miocene pattern.

The magnitude of the dispersion parameters for our collection is comparable to that of the data for the last 5 Ma in which the geomagnetic field reversed frequently (Table 6). On the other hand, data for the Cretaceous superchron (Biggin *et al.* 2008a; Mena *et al.* 2010) support the hypothesis of a correlation between PSV and reversal frequency. PSV analysis can thus be a ffirst-order tool for examining geomagnetic polarity stability in geological times. Interestingly, the few Precambrian results seem to indicate lower PSV than is seen in the Miocene and Plio-Pleistocene. Is this a long-term trend? This merits further investigations. An examination of the palaeomagnetic data bases reveals plenty of Miocene potential targets for PSV determinations.

## ACKNOWLEDGEMENTS

We would like to thank Gregory McCracken for his help in the collecting of the samples during the 2010 fieldwork, Steve Reidel for his guidance and knowledge of the local geology, and the staff at the Institute for Rock Magnetism in the University of Minnesota (Mike Jackson, Julie Bowles, Peter Solheid and Joshua Feinberg) for their help in the rock magnetic analysis. Samantha Nemkin helped with the SEM images. Pavel Doubrovine provided the dispersion calculation program and Andy Biggin helped with the calculation of the error limits on the dispersion parameters and made valuable comments on the 2012 thesis draft of this project. Jonathan Hagstrum and Michael Sawlan of the USGS-Menlo Park made their compilation of CRB data available to us. We also would like to extend our gratitude to the Nez Perce Tribe for granting us permission to sample on their property. The journal's editor, Richard Holme, and reviewers Catherine Johnson and Cor Langereis, are thanked for their constructive and valuable comments that helped improve the manuscript. This study was supported by the Division of Earth Sciences, the U.S. National Science Foundation, grant EAR-0909288, funded under the American Reinvestment and Recovery Act.

## REFERENCES

Atwater, T.M. & Stock, J.M., 1998. Pacific-North America plate tectonics of the Neogene southwestern United States: an update, *Int. Geol. Rev.,* **40,** 375–402.

Barry, T.L., Self, S., Kelley, S.P., Reidel, S., Hooper, P. & Widdowson, M., 2010. New $^{40}Ar/^{39}Ar$ dating of the Grande Ronde lavas, Columbia River Basalts, USA: implications for duration of flood basalt eruption episodes, *Lithos,* **228,** 213–222.

Bazhenov, M.L., Van der Voo, R., Levashova, N.M. & Dominguez, A.R., 2013. Late Devonian paleomagnetism of the North Tien Shan, Kyrgyzstan: can secular variation vary on a short time scale? *Geophys. J. Int.,* **193,** 635–649.

Bazhenov, M.L., Van der Voo, R., Meert, J.G., Levashova, N.M. & Ipatieva, I.S., 2014. Late Paleozoic geomagnetic field estimates from studies of Permian lavas in northeastern Kazakhstan, *Russian Geol. Geophys.,* **55,** 108–117.

Beck, M.E. Jr., 1976. Discordant paleomagnetic pole position as evidence of regional shear in the Western Cordillera of North America, *Am. J. Sci.,* **276,** 694–712.

Beck, M.E. Jr., 1980. Paleomagnetic record of plate-margin tectonic processes along the western edge of North America, *J. geophys. Res.,* **80,** 7115–7131.

Biggin, A.J., Van Hinsbergen, D.J.J., Langereis, C.G., Straathof, C.B. & Deenen, M.H.L., 2008a. Geomagnetic secular variation in the Cretaceous Normal superchron and in the Jurassic, *Phys. Earth planet. Inter.,* **169,** 3–19.

Biggin, A.J., Strik, G. & Langereis, C.G., 2008b. Evidence for a very long-term-trend in geomagnetic secular variation, *Nat. Geosci.,* **1,** 395–398.

Biggin, A.J., Steinberger, B., Aubert, J., Suttie, N., Holme, R., Torsvik, T.H., van der Meer, D.G. & van Hinsbergen, D.J.J., 2012. Possible links between long-term geomagnetic variations and whole-mantle convection processes, *Nat. Geosci.,* 674, doi:10.1038/ngeo1558.

Chamberlain, V.E. & Lambert, R.St.J., 1994. Lead isotopes and the sources of the Columbia River Basalt Groups, *J. geophys. Res.,* **99,** 11 805–11 817.

Choiniere, S.R. & Swanson, D.A., 1979. Magnetostratigraphy and correlation of Miocene basalts of the northern Oregon and Columbia Plateau, southeast Washington, *Am. J. Sci.,* **279**(7), 755–777.

Cogné, J.P., 2003. PaleoMac: a Macintosh (TM) application for treating paleomagnetic data and making plate reconstructions, *Geochem. Geophys. Geosyst.,* **4,** 1–8.

Cox, A., 1970. Latitude dependence of the angular dispersion of the geomagnetic field, *Geophys. J. R. astr. Soc.,* **20,** 253–269.

Cromwell, G., Tauxe, L., Staudigel, H., Constable, C.G., Koppers, A.A. & Pedersen, R.-B., 2013. In search of long-term hemispheric asymmetry in the geomagnetic field: results from high northern latitudes, *Geochem. Geophys. Geosystems,* **14,** doi:10.1002/ggge.20174.

Day, R., Fuller, M. & Schmidt, V.A., 1977. Hysteresis properties of titanomagnetites: grain size and composition dependence, *Phys. Earth planet. Inter.,* **13,** 260–267.

Dominguez, A.R., 2012. *On the behavior of the geomagnetic field during the Miocene, MSc thesis*, University of Michigan, 129 p. Available at: http://hdl.handle.net/2027.42/92475 (last accessed 21 December 2013).

Druken, K.A., Long, M.D. & Kincaid, C., 2011. Patterns in seismic anisotropy driven by rollback subduction beneath the High Lava Plains, *Geophys. Res. Lett.,* **38,** L13310, doi:10.1029/2011GL047541.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. Roy. Soc., London,* **A217,** 295–305.

Fouch, M.J., 2012. The Yellowstone hotspot: plume or not?, *Geology,* **40,** 479–480.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

Franco, D.R., Ernesto, M., Ponte-Neto, C.F., Hinnov, L.A., Berquó, T.S., Fabris, J.D. & Rosière, C.A., 2012. Magnetostratigraphy and mid-paleolatitude VGP dispersion during the Permo-Carboniferous Superchron: results from Paraná Basin (S. Brazil) rhythmites, *Geophys. J. Int.,* **191,** 993–1014.

Haldan, M.M., Langereis, C.G., Biggin, A.J., Dekkers, M.J. & Evans, M.E., 2009. A comparison of detailed equatorial red bed records of secular variation during the Permo-Carboniferous reversed superchron, *Geophys. J. Int.,* **177,** 834–848.

Hooper, P.R., 2000. Chemical discrimination of Columbia River basalt flows, *Geochem. Geophys. Geosys.,* **1,** 1–14.

Hooper, P.R. & Conrey, R.M., 1989. A model for the tectonic setting of the Columbia River Basalt eruptions, *Geol. Soc. Am. Special Paper,* **239,** 293–306.

Jacobs, J.A., 1994. *Reversals of the Earth's Magnetic Field,* 2nd edn, Cambridge Univ. Press, 346 pp.

Jarboe, N.A., Coe, R.S., Renne, P.R., Glen, J.M.G. & Mankinen, E.A., 2008. Quickly erupted volcanic sections of the Steens Basalt, Columbia River Basalt Group: secular variation, tectonic rotation, and the Steens Mountain reversal, *Geochem. Geophys. Geosys,* **9,** 1–24.

Johnson, C.L. *et al.*, 2008. Recent investigations of the 0–5 Ma geomagnetic field recorded by lava flows, *Geochem. Geophys. Geosys.,* **9,** Q04032, doi:10.1029/2007GC001696.

Kirschvink, J.L., 1980. The least squares line and plane and the analysis of paleomagnetic data, *Geophys. J. R. astr. Soc.,* **62,** 699–718.

Kodama, K.P., 2012. *Paleomagnetism of Sedimentary Rocks: Process and Interpretation,* Wiley-Blackwell, 157 pp.

Kono, M., 1997. Distributions of paleomagnetic directions and poles, *Phys. Earth planet. Inter.,* **103,** 313–327.

Laj, C. & Channell, J.E.T., 2007. Geomagnetic excursions, in *Treatise on Geophysics,* Vol. 5, pp. 373–416, ed. Kono, M., Elsevier.

Linder, J. & Gilder, S.A., 2011. Geomagnetic secular variation recorded by sediments deposited during the Cretaceous normal superchron at low latitude, *Phys. Earth planet. Inter.,* **187,** 245–260.

Linder, J. & Gilder, S.A., 2012. Latitude dependency of the geomagnetic secular variation S parameter: a mathematical artifact, *Geophys. Res. Lett.,* **39,** L02308, doi:10.1029/2011GL050330.

Long, P.E. & Duncan, R.A., 1983. $^{40}Ar/^{39}Ar$ ages of Columbia River Basalts from deep boreholes in south-central Washington, *EOS, Trans. Am. geophys. Un.,* **64**(9), 90.

Long, M.D. *et al.*, 2012. Mantle dynamics beneath the Pacific Northwest and the generation of voluminous back-arc volcanism, *G-cubed,* **13,** doi:10.1029/2012GC004189.

Lowrie, W. & Kent, D.V., 1983. Geomagnetic reversal frequency since the Late Cretaceous, *Earth planet. Sci. Lett.,* **62,** 305–313.

Lowrie, W. & Kent, D.V., 2004. Geomagnetic polarity timescales and reversal frequency regimes, in *Timescales of the Paleomagnetic Field,* Vol. 135, pp. 117–129, eds Channell, J.E.T., Kent, D.V., Lowrie, W. & Meert, J.G., Geophys. Monogr. Ser. AGU.

Magill, J.R., Wells, R.E., Simpson, R.W. & Cox, A.V., 1982. Post 12 m.y. rotation of southwest Washington, *J. geophys. Res.,* **87,** 3761–3776.

Mangan, M.T., Wright, T.L., Swanson, D.A. & Byerly, G.R., 1986. Regional correlation of Grande Ronde Basalt flows, Columbia River Basalt Group, Washington, Oregon, and Idaho, *Bull. geol. Soc. Am.,* **97**(11), 1300–1318.

McElhinny, M.W. & McFadden, P.L., 1997. Palaeosecular variation over the past 5 Myr based on a new generalized database, *Geophys. J. Int.,* **131**(2), 240–252.

McElhinny, M.W. & McFadden, P.L., 2000. *Paleomagnetism—Continents and Oceans,* Academic Press, pp. 386.

McFadden, P.L. & McElhinny, M.W., 1988. The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth planet. Sci. Lett.,* **87,** 161–172.

McFadden, P.L. & McElhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

McFadden, P.L., Merrill, R.T. & McElhinny, M.W., 1988. Dipole quadrupole family modeling of paleosecular variation, *J. geophys. Res.,* **93**(B10), 11 583–11 588.

McFadden, P.L., Merrill, R.T., McElhinny, M.W. & Lee, S.H., 1991. Reversals of the Earth's magnetic-ffield and temporal variations of the dynamo families, *J. geophys. Res.,* **96**(B3), 3923–3933.

McKee, E.H., Hooper, P.R. & Kleck, W.D., 1981. Age of Imnaha basalt; oldest flows of the Columbia River Basalt Group, *Northwest United States: Isochron/West.,* **31,** 31–33.

McKee, E.H., Swanson, D.A. & Wright, T.L., 1977. Duration and volume of Columbia River basalt volcanism; Washington, Oregon, and Idaho, *Geol. Soc. Am. Abstracts with Programs,* **9,** 463.

Mena, M., Goguitchaichvili, A., Cervantes Solano, M. & Vilas, J.F., 2010. Paleosecular variation and absolute geomagnetic paleointensity records retrieved from the Early Cretaceous Posadas Formation (Misiones, Argentina), *Studia Geophysica et Geodaetica,* **55,** 279–309.

Menzo, Z., 2013. *Permian secular variation of the Ayaguz formation in Kazakhstan, Honors thesis,* University of Michigan. Available at: http://hdl.handle.net/2027.42/97753 (last accessed 21 December 2013).

Olsen, N. & Mandea, M., 2007. Will the magnetic north pole move to Siberia? *EOS, Trans. Am. geophys. Un.,* **88,** 293–294.

Reidel, S.P., 1978. *Stratigraphy and petrogenesis of the Grande Ronde Basalt in the lower Salmon and adjacent Snake River Canyons, PhD thesis*, Washington State Univ., Pullman, Washington, 415 p.

Reidel, S.P., 1983. Stratigraphy and petrogenesis of thhe Grande Ronde Basaltr from the deep canyon country of Washington, Oregon and Idaho, *Bull. geol. Soc. Am.,* **94,** 519–542.

Reidel, S.P., Martin, B.S. & Petcovic, H.L., 2003. The Columbia River flood basalts and the Yakima fold belt, in *Western Cordillera and Adjacent Areas,* Vol. 4, pp. 87–105, ed. Swanson, T.W., Geological Society of America Field Guide.

Reidel, S.P., Long, P.E., Myers, C.W. & Mase, J., 1981. New evidence for greater than 3.2 kilometers of Columbia River basalt beneath central Columbia Plateau (abstract), *EOS, Trans. Am. geophys. Un.,* **63,** 173.

Reidel, S.P., Tolan, T.L., Hooper, P.R., Beeson, M.H., Fecht, K.R., Bentley, R.D. & Anderson, J.L., 1989. The Grande Ronde Basalt, Columbia River Basalt Group: stratigraphic descriptions and correlations in Washington, Oregon, and Idaho, in *Volcanism and Tectonism in the Columbia River Flood-Basalt Province,* Vol. 239, pp. 21–53, eds Reidel, S.P. & Hooper, P.R., Geological Society of America Special Paper.

Sheriff, S.D. & Bentley, R.D., 1980. Anomalous paleomagnetic directions from the Miocene Wanapum Basalt, Washington and Oregon (abstract), *EOS, Trans. Am. geophys. Un.,* **62,** 949.

Simpson, R.W., Magill, J.R., Wells, R.E. & Cox, A.V., 1980. Post 12 m.y. rotation in southwest Washington (abstract), *EOS, Trans. Am. geophys. Un.,* **61,** 949.

Smirnov, A.V., Tarduno, J.A. & Evans, D.A.D., 2011. Evolving core conditions ca. 2 billion years ago detected by paleosecular variation, *Phys. Earth planet. Inter.,* **187,** 225–231.

Swanson, D.A., Anderson, J.L., Bentley, R.D., Byerly, G.R., Camp, V.E., Gardner, J.N. & Wright, T.L., 1979a, Reconnaissance geologic map of the Columbia River Basalt Group in parts of eastern Washington and northern Idaho, *Open-File Rep, U.S. Geol. Surv.,* 79–1363, scale 1:250,000, 12 sheets.

Swanson, D.A., Wright, T.L., Hooper, P.R. & Bentley, R.D., 1979b, Revisions in strati- graphic nomenclature of the Columbia River Basalt Group, *U.S.: Geol. Surv. Bull.,* **1475-G,** G54–G59.

Tauxe, L., 2010. *Essentials of Paleomagnetism, with Contributions From,* pp. 489, eds Banerjee, S.K., Butler, R.F. & Van der Voo, R., Univ. California Press.

Tauxe, L. & Kent, D.V., 2004. A simplified statistical model for the geomagnetic field and the detection of shallow bias in Paleomagnetic inclinations: was the ancient magnetic field Dipolar?, *Timescales of the Paleomagnetic Field, Geophys. Monogr. Ser.,* Vol. 145, pp. 101–115, AGU.

Tauxe, L. & Kodama, K.P., 2009. Paleosecular variation models for ancient times: clues from Keweenawan lava flows, *Phys. Earth planet. Inter.,* **177,** 31–45.

Tauxe, L. & Watson, G.S., 1994. The fold test; an eigen analysis approach, *Earth planet. Sci. Lett.,* **122**(3–4), 331–341.

Tolan, T.L., Reidel, S.P., Beeson, M.H., Anderson, J.L., Fecht, K.R. & Swanson, D.A., 1989. Revisions to the estimates of the areal extent and volume

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016

of the Columbia River Basalt Group, in *Volcanism and Tectonism in the Columbia River Flood-Basalt Province,* Vol. 239, pp. 1–20, eds Reidel, S.P. & Hooper, P.R., Geological Society of America Special Paper.

Tolan, T.L., Martin, B.S., Reidel, S.P., Anderson, J.L., Lindsey, K.A. & Burt, W., 2009. An Introduction to the stratigraphy, structural geology, and hydrogeology of the Columbia River Flood-Basalt Province: a primer for the GSA Columbia River Basalt trips, *Geol. Soc. Am. Field Guide,* **15,** 599–643.

Vandamme, D., 1994. A new method to determine paleosecular variation, *Phys. Earth planet. Inter.,* **85,** 131–142.

Waters, A.C., 1961. Stratigraphic and lithologic variations in the Columbia River basalt, *Am. J. Sci.,* **259,** 583–611.

Watkins, N.E. & Baksi, A.K., 1974. Magnetostratigraphy and oroclinal folding of the Columbia River, Steens and Owyhee basalts in Oregon, Washington and Idaho, *Am. J. Sci.,* **274,** 148.

Wells, R.E., Simpson, R.W., Bentley, R.D., Beeson, M.H., Mangan, M.T. & Wright, T.L., 1989. Correlation of Miocene flows of the Columbia River Basalt Group from the central Columbia River Plateau to the coast of Oregon and Washington, *Geol. Soc. Am. Special Paper,* **239,** 13–129.

Zijderveld, J.D.A., 1967. *A.C. Demagnetization of Rocks: Methods in Palaeomagnetism,* pp. 256–286, eds Collinson, D.W., Creer, K.M. & Runcorn, S.K., Elsevier.

## SUPPORTING INFORMATION

Additional Supporting Information may be found in the online version of this article:

**Table S1.** Palaeomagnetic site means, site-numerical order, grouped, N only.
**Table S2.** Palaeomagnetic site means, site-numerical order, grouped, R only.
**Table S3.** Published Plateau Directions & VGP's.
**Table S4.** Radiometric age dates, methodology, references.
**Table S5.** The eight criteria, for use in PSV studies, from McElhinny & McFadden (1997) ('MM97')
(http://gji.oxfordjournals.org/lookup/suppl/doi:10.1093/gji/ggt487/-/DC1)

Please note: Oxford University Press is not responsible for the content or functionality of any supporting materials supplied by the authors. Any queries (other than missing material) should be directed to the corresponding author for the article.

Downloaded from http://gji.oxfordjournals.org/ at University of Western Australia on October 13, 2016